

# MANDATE-primary 2026Q2 Empirical Evaluation: Empirical Evidence Supplemental

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## Abstract

This supplement provides the empirical evidence supporting the manuscript’s claims about MANDATE-primary, a six-role fine-tuned LLM specification pipeline. It compiles the methodology, dataset construction, substantive findings catalog with verbatim model output samples, cross-system structural comparison against six baselines, and the structured Preregistration Deviation Table following Willroth & Atherton (2024). Five of six MANDATE roles exhibit content-tripwire failure modes characterized at Phase 6 scale across 1500 MANDATE-primary RunRecords and 7536 baseline RunRecords (7236 main-matrix records across six baselines plus 300 B1 hold-out records). The v1 Phase 8 three-judge stratified-sample grading executed the full pre-registered protocol; PROTOCOL\_LOCK §8 engaged the v2 supersession path, under which the v2 cycle introduces a shape-neutral rubric matched to canonical MANDATE’s free-form-object schema (documented in the companion **v2 Protocol Amendment**). Project-internal engineering chronology and patch history are documented separately in **Engineering and Operational Provenance**, also bundled with the deposit.

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# 1 Executive Overview

**Document scope.** This supplement compiles the v1 empirical evidence supporting the manuscript’s claims about MANDATE-primary against the `mandate-eval-primary-2026q2-v1` freeze tag (commit `4f8af83`). All nine v1 deviations from pre-registration are recorded in the structured Deviation Table (Section 9) following Willroth & Atherton (2024). The five content-tripwire findings (Section 4) are evidenced at full coverage from on-disk RunRecord inspection.

**v1 → v2 rubric refinement.** v2 introduces a shape-neutral grading rubric matched to canonical MANDATE’s free-form-object schema, alongside a three-condition measured design (Cond-X regrade of v1 outputs under the v2 rubric, Cond-A canonical MANDATE on pre-extracted structured input, Cond-B canonical MANDATE with LLM-augmented Interpreter end-to-end). The v1 cross-system numerical row in Section 5.4 documents the v1 rubric measurement against the v1 ground-truth scaffolds; v2 is the source of cross-system claims for citation. v2 methodology lives in the companion **v2 Protocol Amendment**; engineering chronology and patch history live in **Engineering and Operational Provenance**, both bundled with the deposit.

The 2026Q2 evaluation produces 9036 RunRecords across seven systems on a 120-task source-conditioned main corpus and a 30-task out-of-domain hold-out. The principal substantive findings are:

1. **Decomposition single-COA prior reaffirmed at scale.** Across 1480 MANDATE-primary records (1180 successful main matrix + 300 hold-out), every record produces *exactly one* course of action. The canonical name (**Minimal manual assessment approach**) and two-node DAG shape (Manual Assessment → Results Documentation) are stable across all measured variation.
2. **Interpreter content-tripwire mode flip.** The Interpreter role’s `anchor.minimum.description` falls into one of two patterns: clean structured distillation or deterministic-prefix paragraph echo. Which mode is selected by chunk content fed to the upstream Procedure-role retriever, not by the user-facing request paragraph. Per-domain clean rates: 64.5% FIN (258/400), 56.0% INT (224/400), 76.3% SEC (290/380; SEC denominator is 380 not 400 because the 20 Intake content-tripwire failures are upstream of Interpreter and do not enter the Interpreter mode classification).
3. **Validator gap-acknowledgment instability.** The Validation role flags `anchor-vs-COA-count` gaps via heuristic phrases on 14–19% of records, inconsistently across otherwise-similar inputs.
4. **Binding structured refusal.** The Binding role emits `{"error": "..."}`  when receiving contradictory inputs. Refusal rates: 2.0% FIN, 40.2% INT, 13.5% SEC, 7.0% out-of-domain hold-out. The INT rate inverts the demo-era prediction that financial content would dominate the refusal distribution.
5. **Intake content-tripwire.** On two tasks containing the natural-language phrase “Here’s the constraint:” followed by complex English, the `mandate-intake` fine-tune emits the English sentence as a constraint string, which fails the constraint grammar validator. 20 of 1200 main records (1.7%) affected; reproducible across all 10 seeds per task. Of 8 main-corpus tasks derived from the same source document (NIST SP 800-115), only the 2 carrying the trigger phrase fail.

The combined picture demonstrates that fine-tuned multi-role pipelines exhibit content-sensitive failure modes whose detection at scale requires a structured measurement apparatus and grounded

ground truth. Five of six MANDATE roles show measurable content-tripwire behavior in v1-grounded evaluation. Section 5 quantifies the structural differences between MANDATE-primary’s mandate-as-code output and six baselines’ specification schemas; Section 6 documents the apparatus engineering required to measure these phenomena reliably.

**Deposit state at v1 close.** The `outputs_freeze_v1_1` tag at commit `5f4de54` carries the 9036 RunRecords described in this document. The five-tag freeze chain (corpus, baseline, ground truth, perturbation, outputs) preserves the provenance integrity required for re-evaluation. Phase 8 three-judge grading executed the full pre-registered stratified-sample protocol on 700 records; PROTOCOL\_LOCK §8 engaged the v2 supersession path on schema-sensitive subjective outcomes (quantified in Section 4.6). The v2 deposit will carry the `mandate-eval-primary-2026q2-v2` tag and is the source of cross-system claims for citation; v1 evidence in this document remains the v1 record of measurement.

## 1.1 Relationship to Prior Published Work

The MANDATE framework was first reported in Calboreanu (2026, in review at Frontiers in AI; prior submissions desk-rejected at Requirements Engineering). That paper introduced the six-role decomposition, formalized the gap-detection mechanism, and reported a single-system pilot evaluation of the AEGIS reference implementation across eight paper-derived scenarios. Reviewers across two Frontiers submissions converged on a single substantive complaint: the empirical evaluation was preliminary, self-referential, and lacked comparison. The 2026Q2 evaluation documented in this supplement is the explicit corrective response to that feedback.

**Pre-registration provenance.** The 2026Q2 protocol was locked at pre-registration deposit prior to any 2026Q2 data collection. The locked package lives at `eval_package/` (sibling to the published paper repository) and comprises the master playbook (`00_PLAYBOOK_v2.md`), pre-registration template (`00_PREREGISTRATION_TEMPLATE.md`), analysis plan (`ANALYSIS_PLAN.md`), the locked prompts (`PROMPTS.md`), evaluation forms (`FORMS.md`), the six hand-authored calibration tasks (`calibration_tasks/cal_01..06.json`), and the canonical `PROTOCOL_LOCK.md` that owns the  $\kappa \geq 0.40$  halt rule applied in v1 Phase 8 (§5.4 below). All design decisions in this supplement defer to the locked package; deviations are catalogued verbatim in Section 9.

**Reviewer-objection-to-component map.** Each substantive complaint raised in the two Frontiers in AI rejection letters maps to a specific 2026Q2 design choice. The map below is the bridge between the published paper’s preliminary evidence and this supplement’s headline corpus.

| Reviewer<br>(paraphrased<br>response_to_reviewers.pdf)                 | objection<br>from | 2026Q2 design response                                                                                                                     | Where in this supplement |
|------------------------------------------------------------------------|-------------------|--------------------------------------------------------------------------------------------------------------------------------------------|--------------------------|
| R1: “Empirical work preliminary”<br>(8 scenarios × 1 backend in paper) |                   | 1500-record main matrix × 7 systems × 10 seeds source-conditioned corpus                                                                   | §3, §4, §5               |
| R3: “Absence of comparative evaluation”                                |                   | 6 pre-specified baselines (B1–B6) + three-judge ensemble grading                                                                           | §2.2, §5                 |
| R3: “Self-referential validation”                                      |                   | Source-conditioned corpus authoring from 18 public documents (NIST, COSO, PCAOB, MITRE, ODNI, etc.)                                        | §2.1                     |
| R3: “Robustness not demonstrated”                                      |                   | 350-perturbation suite (7 types × 50 trials each)                                                                                          | §7                       |
| R3: “Scalability not demonstrated”                                     |                   | 154-hour Ollama serial wall clock + per-domain stratification                                                                              | §3.3                     |
| R3: “No out-of-domain generalization”                                  |                   | 30-task <b>software_engineering</b> hold-out (4th domain)                                                                                  | §3.4                     |
| R3: “Validation against rule-based system only”                        |                   | Calibration tasks + model-authored ground-truth scaffolds (SME ratification pending, Deviation 1) + three independent vendor-family judges | §2.3, §2.4               |
| R1: “Limited domain coverage” (1 domain in paper)                      |                   | 3 in-domain (security ops, financial, intelligence) + 1 hold-out                                                                           | §2.1, §3.4               |
| R1: “No pre-registration”                                              |                   | Full pre-registration deposit at <b>eval_package/</b> locked before data collection                                                        | §1.1 (this section)      |
| R3: “Single-vendor LLM execution” (Qwen3 only)                         |                   | Multi-vendor Cond-B at stratified-300 scale across four LLM vendor families                                                                | v2 Protocol Amendment §3 |

**v0 pilot baseline from Calboreanu (2026), Section 12.** The published paper reports three substantive empirical tables. They are reproduced verbatim below as the v0 single-system baseline this supplement extends. Cross-references in the table-footnote markers (<sup>a</sup>, <sup>b</sup>, <sup>c</sup>) and the metric definitions are stable across the published paper and this supplement.

Table 1: v0 Static Evaluation Results from Calboreanu (2026, §12). AEGIS reference implementation, 500-test deterministic suite.

| #  | Metric                    | Result          | Notes                                           |
|----|---------------------------|-----------------|-------------------------------------------------|
| 1  | Test suite pass rate      | 499/500 (99.8%) | 1 skip (known)                                  |
| 2  | Anchor field extraction   | 7/7 (100%)      | All M,T,C fields present                        |
| 3  | Gap detection precision   | 96.8% (30/31)   | Structurally complete examples                  |
| 4  | Gap detection recall      | 96.8% (30/31)   | 33 examples excluded (input-only corpus design) |
| 5  | Trace chain integrity     | 2/2 (100%)      | All hashes verify                               |
| 6  | Anchor hash (Algorithm 1) | 1/1 (100%)      | Production hashing module                       |
| 7  | COA diversity             | 3/3 (100%)      | Structural variation confirmed                  |
| 8  | Cross-domain pipeline     | 5/5 (100%)      | Generic, IR, Intel, Pentest, Underspec          |
| 9  | Constraint grammar        | 12/12 (100%)    | All grammar patterns parsed                     |
| 10 | NIST AI RMF mapping       | 11/11 (100%)    | All MAP & MEASURE subcategories                 |
| 11 | Registry matching         | 5/5 (100%)      | All match types validated                       |
| 12 | Readiness scores          | 1/1 (100%)      | 17% (1/6), blocking=True                        |
| 13 | Timing (8 scenarios)      | <100ms total    | Deterministic, no LLM required                  |

Table 1 establishes that the deterministic implementation passes all structural mechanisms (gap detection, constraint grammar, anchor hashing, trace integrity, registry matching, NIST AI RMF mapping) at near-perfect rates on the static suite. This is the prior baseline against which the

2026Q2 evaluation adds LLM-mode variance, cross-system comparison, and stratified corpus generalization.

Table 2: v0 LLM-Backed Pipeline Execution from Calboreanu (2026, §12). Fine-tuned Qwen3 models via Ollama, 8 paper-derived scenarios.

| #             | Scenario           | Status  | LLM   | Fallback | Notes                            |
|---------------|--------------------|---------|-------|----------|----------------------------------|
| 1             | CISO Report        | SUCCESS | 4/6   | 2/6      | Intake timeout, Validation parse |
| 2             | Undef. Minimum     | GAP_RPT | 5/6   | 1/6      | Validation parse error           |
| 3             | Undef. Target      | GAP_RPT | 5/6   | 1/6      | Interpreter timeout              |
| 4             | Unknown Pattern    | GAP_RPT | 6/6   | 0/6      | Full LLM execution               |
| 5             | Missing Capability | GAP_RPT | 6/6   | 0/6      | Full LLM execution               |
| 6             | Unassess. Risk     | GAP_RPT | 5/6   | 1/6      | Decomposition parse              |
| 7             | Ransomware IR      | SUCCESS | 4/6   | 2/6      | Intake timeout, Validation parse |
| 8             | OSINT Intel        | GAP_RPT | 5/6   | 1/6      | Validation parse error           |
| <i>Totals</i> |                    |         | 40/48 | 8/48     | 83% LLM, 17% fallback            |

Table 2 reports the v0 LLM-mode pilot: across 48 role invocations (8 scenarios  $\times$  6 roles), 40 (83%) executed via LLM and 8 (17%) fell back to the deterministic path under fine-tuned Qwen3 (Intake and Procedure on Qwen3-8B, the remaining four roles on Qwen3-32B, all via Ollama on Mac mini M4 Pro). All 8 scenarios completed successfully regardless of execution mode.

Table 3: v0 Structural Property Comparison: Deterministic vs. LLM Execution from Calboreanu (2026, §12).

| Scenario          | Status                           | Hash     | Gaps     | COAs     | Trace |
|-------------------|----------------------------------|----------|----------|----------|-------|
| 01 CISO Report    | ✓                                | ✓        | ✓        | ✓        | ✓     |
| 02 Undef. Min     | ✓                                | $\neq^a$ | $\neq^b$ | ✓        | ✓     |
| 03 Undef. Tgt     | ✓                                | $\neq^a$ | $\neq^b$ | ✓        | ✓     |
| 04 Unknown Pat    | ✓                                | $\neq^a$ | $\neq^b$ | ✓        | ✓     |
| 05 Missing Cap    | ✓                                | $\neq^a$ | $\neq^b$ | ✓        | ✓     |
| 06 Unassess. Risk | ✓                                | $\neq^a$ | $\neq^b$ | ✓        | ✓     |
| 07 Ransomware IR  | ✓                                | $\neq^a$ | ✓        | $\neq^c$ | ✓     |
| 08 OSINT Intel    | SUCCESS $\rightarrow$ GAP_REPORT | $\neq^a$ | $\neq^b$ | $\neq^c$ | ✓     |

<sup>a</sup>Anchor hashes differ because LLM extracts richer content; Property 1 holds within each run.

<sup>b</sup>LLM detects additional gaps beyond deterministic baseline (more thorough analysis).

<sup>c</sup>LLM Decomposition makes different strategic COA judgments.

**Key finding (paper):** Trace completeness (P2) holds universally: 6-entry chains on every run regardless of execution mode.

Table 3 is the v0 structural-property finding the 2026Q2 evaluation generalizes. Trace completeness holds across all 8 scenarios under both deterministic and LLM execution, which was the paper’s headline structural-invariance claim. The 2026Q2 extension tests whether this generalization survives at the scale of 1500 MANDATE-primary records on source-conditioned tasks the fine-tunes have not seen.

**Evidence-tier chain.** The published Section 12 evidence, the April 2026 cross-profile pilot reported in §6.7, and the 2026Q2 main matrix form a single chain of increasing scale and rigor. Table 4 summarizes.

Table 4: Evidence-Tier Chain: Prior Pilot → 2026Q2 Main Matrix

| Tier                     | Time    | Scope                              | Systems                            | Records             |
|--------------------------|---------|------------------------------------|------------------------------------|---------------------|
| v0 (paper §12)           | 2026Q1  | 8 paper-derived scenarios          | 1 (Qwen3 fine-tunes)               | 48 role invocations |
| v0.5 (April pilot, §6.7) | 2026-04 | 6-case <code>authorized_lab</code> | 3 (det / Qwen3-base / Qwen3-tuned) | 15 logged runs      |
| v1 (this evaluation)     | 2026Q2  | 120 main + 30 hold-out             | 7 (MANDATE-primary + B1–B6)        | 9036 RunRecords     |
| v2 (forward)             | 2026Q3+ | + multi-vendor Cond-B              | 7+ (with planned cross-vendor)     | + Cond-A/B records  |

The full per-artifact routing between this supplement, the GitHub replication package, and the Zenodo archive is documented in `DEPOSIT_MAPPING.md` (sibling file). The Zenodo package structure is documented in `GITHUB_DEPOSIT_PLAN.md`. Both mapping documents are mirrored into the deposit at submission.

## 1.2 What This Evaluation Demonstrates

The combined v0 (Calboreanu 2026, §12), v0.5 (April 2026 cross-profile pilot, §6.7), and v1 (2026Q2 main matrix, §3–§4) evidence supports five independent affirmative claims about MANDATE. Each claim is backed by on-disk evidence that does not depend on inter-judge  $\kappa$  stability, rubric-shape choices, or LLM-as-judge methodology. Stage 4 v2 grading (completed 2026-07-01) adds the comparative evaluation tier reviewers requested, but the five claims below stand on the structural counts and verbatim model outputs alone.

**Claim 1: MANDATE produces structurally valid mandate-as-code end-to-end at scale.** “Structurally valid” here is defined as: the apparatus produces a complete mandate-as-code artifact (anchor, `courses_of_action`, trace chain of length  $\geq 6$ , recommendation, `gap_reports`, `registry_reference`) that validates against the canonical JSON schema (`mandate-as-code.schema.json`) and whose trace chain hashes resolve. Structural validity is necessary but not sufficient for output quality: substantive content correctness is addressed in §4 (Findings) and §5.4 (cross-system scoring), with scope-of-claim caveats enumerated in Section 11. The claim made here is that the apparatus produces well-formed output the quality-grading stage can score in the first place. Across the 2026Q2 main matrix, every Cond-A record (1500/1500) and every Cond-B record (1500/1500) terminated with `ok=True`, producing the required anchors, COA structure, trace chain, and gap report. Across the 1500 v1 MANDATE-primary records, **1480/1500 (98.7%)** produce a valid mandate-as-code artifact; the 20-record delta is the documented Intake content-tripwire failure mode on the two SEC tasks carrying the trigger phrase “Here’s the constraint:” (TASK-MAIN-SEC-038 and TASK-MAIN-SEC-040, 10 seeds each), characterized in detail in Section 4.5 and counted in §1.1’s reviewer-objection map. The 20 deltas are reproducible failures the measurement apparatus surfaced at scale (exactly the type of fix-targeted signal MANDATE is designed to expose), and they are documented openly rather than masked. The published paper’s static-evaluation suite reports a 499/500 (99.8%) pass rate on the mandate-specific test slice (Table 1). The broader full AEGIS-eval test suite (covering MANDATE plus the surrounding governance harness) reports 1,426 / 28 / 5 (passed / skipped / failed) on the v2-deposit codebase; the 5 failures are FastAPI endpoint integration tests hitting sandbox-specific filesystem permission errors on evidence-file unlink, and are not formal-property regressions. The independent MLT-Governance-Stack v1.0.0rc1 canonical implementation reports 1,942 collected tests across 116 engine modules. This is the floor under every downstream claim: the apparatus does not crash, hang, or produce malformed output on real public-source-derived tasks, and three independent test corpora cross-verify the formal-property mechanisms.



**Claim 2: MANDATE detects specification defects with high precision and recall on planted probes.** The published paper’s gap-detection results report 96.8% precision (30 true positives, 1 false positive) and 96.8% recall (30 of 31 resolvable cases) on the 31-example *structurally complete subset*, plus 37/37 dedicated gap-detection tests passing (Table 1, rows 3–4 and supporting text). An additional 33 gap-producing examples in the corpus contain only input payloads without embedded runtime output, making automated gap detection structurally impossible for those cases by corpus-design; the published paper transparently discloses that these 33 are excluded from the structurally-complete recall calculation rather than masked as false negatives. The 96.8%/96.8% number is the apparatus’ performance on what it can reach by design; the 33 input-only examples are a corpus-design boundary, not a detection failure. Gap detection catches omissions, underspecified thresholds, and contradictions in mission inputs on structurally complete inputs; the corpus-design boundary of the 33 input-only examples is documented rather than elided.

**Claim 3: MANDATE’s structural invariants hold across execution modes and LLM vendor families.** The evidence base for this claim spans four independent axes:

- **Deterministic vs LLM (paper §12).** The published paper’s comparison (Table 3) reports trace completeness (Property 2) holding universally across all 8 scenarios under both deterministic and LLM-augmented execution.
- **Cross-profile (April 2026 pilot, §6.7).** The same ok-rate (5/6) across deterministic, Qwen3-base, and Qwen3-tuned configurations on the 6-case `authorized_lab` corpus: LLM augmentation neither increases nor degrades pipeline completion.
- **v1 main matrix at scale.** 1500-record Cond-A and Cond-B both at 1500/1500 structurally valid; Cond-B uses canonical MANDATE with LLM-augmented Interpreter end-to-end.
- **Cross-LLM-family Cond-B, with per-vendor fallback disclosure.** Stratified-300 Cond-B runs against four LLM vendor families via local Ollama produce 1,200/1,200 structurally valid records with 100% trace-chain completeness. The per-vendor, per-domain LLM-fallback rate is however not uniform:

Table 5: Per-vendor per-domain any-role LLM-fallback rate on the cross-vendor Cond-B pilot ( $n = 100$  per cell; 300 records per vendor across FIN/INT/SEC). “Fallback” means at least one of the six roles’ LLM path fell back to its deterministic implementation on the record (union across roles); Interpreter-only schema-validation failure rates are lower (Llama 64%, Phi-3 20%), and Qwen’s 16% SEC cell consists entirely of Procedure-role reference-grounding rejections with zero Interpreter schema failures. Only Qwen carries the LLM path on essentially all records; Llama and Phi-3 rely on the deterministic fallback in at least one role on every record.

| Vendor         | FIN  | INT  | SEC  |
|----------------|------|------|------|
| Qwen 2.5 (32B) | 0%   | 0%   | 16%  |
| Llama 3.2 (3B) | 100% | 100% | 100% |
| Mistral (7B)   | 4%   | 96%  | 100% |
| Phi-3 (14B)    | 100% | 100% | 100% |

On Llama 3.2 and Phi-3, the any-role LLM-fallback union reaches 100% of records (Interpreter-only schema-validation failures: Llama 64%, Phi-3 20%; the remainder of the union comes from the other roles); the apparatus’ deterministic fallback path produces the structurally valid output on those records. The cross-vendor structural completeness therefore demonstrates *graceful*

*degradation when a role’s LLM path fails: MANDATE’s deterministic fallback delivers a schema-valid record across LLM capability tiers rather than LLM-role invariance across vendor families.* Vendors where the LLM paths do carry records end-to-end (Qwen 2.5 primarily; Mistral on FIN) confirm structural completeness also holds when the LLM is the acting extractor. The two mechanisms together, LLM execution when it succeeds and deterministic fallback when it does not, constitute the defense-in-depth guarantee documented in Section 4.4.

The framework is the controlled variable across this evidence chain. Structural completeness of the mandate-as-code output survives execution-mode substitution because MANDATE’s apparatus provides two paths (LLM-augmented and deterministic) that both produce schema-conformant output.

**Claim 4: MANDATE surfaces consequential governance signals in real systems.** Four pieces of direct evidence:

- **AEGIS specification audit (Calboreanu 2026, §12).** 51 prompt-specification defects identified across 9 audit rounds, converging from 15 findings in round 1 to zero in round 9. Cross-lane data-contract mismatches causing silent runtime failures (9.8% of findings) became visible only through MANDATE’s multi-document tiered enforcement; single-pass review missed them.
- **Binding structured refusal at scale (Section 4.4).** Under contradictory inputs (e.g., anchor demands  $\geq 3$  COAs when Decomposition emitted 1) the `mandate-binding` fine-tune emits a valid JSON object with an `error` field containing a natural-language identification of the contradiction (see §4.4 for verbatim samples). The apparatus’ JSON-schema validator then catches that the schema-required `decision_summary` field is absent from the LLM output and activates the deterministic- fallback path. The full mechanism (LLM-level semantic refusal plus schema-level structural rejection plus deterministic fallback) is a defense-in-depth chain: the LLM recognizes the contradiction and refuses to fabricate; the schema validator independently catches the missing required field; the deterministic fallback produces a governance-consumable signal rather than silently binding the contradiction. Per-domain rates of this refusal-cascade behavior (2.0% FIN, 40.2% INT, 13.5% SEC, 7.0% hold-out) demonstrate that all three layers fire together at scale. The v2 candidate patch (`standalone data results/v2_patch/`) routes the LLM’s `error` payload directly to `output.gap_reports` as a first-class gap event, eliminating the fallback step in v2. A system that always produces output regardless of input coherence would be unsuitable for deployment in governance pipelines that depend on downstream auditability.
- **Intake content-tripwire reproducibility (Section 4.5).** A natural-language phrase (“Here’s the constraint:”) triggers a reproducible 10/10-seed failure mode on 2 of 1200 main-corpus tasks; 6 of 8 NIST SP 800-115-derived tasks pass cleanly. The apparatus surfaces a real, characterizable, fix-targeted failure mode at 1.7% prevalence at full corpus scale.
- **Decomposition single-COA prior at scale (Section 4.1).** Across 1480 records, every record produces a single COA with stable name and DAG shape under unstructured input. Identifying this prior required 1480-record scale: no smaller study would have surfaced it. This is the apparatus measuring a structural prior, not a defect.

**Claim 5: MANDATE generalizes across domains.** Three in-domain corpora (`security_operations_reporting`, `financial_reporting`, `intelligence_collection_tasking`) plus a 30-task out-of-domain hold-out (`software_engineering_specification`) all produce

structurally valid output through the same 7-role pipeline. The published paper’s AEGIS implementation spans five operational domains (penetration testing, incident response, multi-cloud security assessment, defense intelligence collection, financial reporting). The April 2026 cross-profile pilot tested the pentest domain under three LLM configurations. Generalization is not a single-corpus artifact.

**What Stage 4 adds.** The Stage 4 full-coverage v2 grading (completed 2026-07-01) produces the comparative-evaluation tier reviewers asked for: MANDATE-primary (Cond-X regrade), Cond-A (pre-extracted), Cond-B (LLM-augmented end-to-end), and B1–B6 baselines all graded under a shape-neutral rubric. Stage 4 closes the comparative objection. It is not the foundation of the affirmative case; Claims 1–5 above are. The completed comparison (Section 5.4) reports that case in its honest form: under the shared raw-text input regime, Cond-B is coverage-competitive with, not dominant over, the strongest agentic baseline, and the differentiating contribution is structural: trace completeness, refusal under contradiction, and cross-vendor structural validity.

**Empirical finish line.** Of the four work-streams that carry the v2 deposit to full pre-registration coverage on every empirically measurable component, three are complete and the fourth is partially complete under documented scope deviations:

- **Full-coverage three-judge grading** on the entire 12,000-record anonymized corpus (Cond-X regrade, Cond-A, Cond-B, and baselines B1–B6) under the v2 shape-neutral rubric. Closes Reviewer R3 “absence of comparative evaluation.”
- **Cross-LLM-family Cond-B** at stratified-300 scale across four vendor families (Qwen 2.5, Llama 3.2, Mistral, Phi-3) via local execution. Closes R3 “single-vendor LLM execution.”
- **Primary ablation A3 (no\_gap\_analysis)** at 1500-record scale plus the apparatus-level invariant characterization documented in Section 4.7 from the attempted A5 ablation. Closes PROTOCOL\_LOCK §5 primary ablation coverage.
- **O5 adversarial perturbation evaluation** on the 350-perturbation suite. Phase A (MANDATE-primary, Cond-A, Cond-B) completed 2026-07-01 with 4,200 records and structural-invariance results reported in Section 7; the structural component of R3 “robustness not demonstrated” closes on Phase A alone. Phase B (baseline perturbation generation, scoped to B1–B4 under Deviation D-12, plus semantic v2 grading of the 18,200-record scoped set) is partially complete: semantic grading was paused 2026-07-08 at 14,685/18,200 main-pass grades (80.7%) under Deviation D-13, with baseline\_4 generation halted at 86.3% at the same closeout and perturbation grading complete for all three MANDATE conditions (4,200/4,200). The cross-system *semantic* comparison of adversarial resistance is deferred to a resumable re-grade; no Phase B semantic claims are made in this supplement.

With the first three work-streams complete and the fourth’s Phase B paused under Deviations D-12 and D-13, the deposit carries on-disk evidence for every empirically measurable pre-registration component except the Phase-B semantic comparison, which is resumable from frozen records. Two remaining open items, namely the source-level ablations (A1, A2, A4, A6, A7) requiring upstream changes per the manifest’s AEGIS\_VARIANT classification and the 30-specification human expert upper bound subject to practitioner availability, are disclosed as external-dependency-bound deviations in Section 9. This is the empirical floor at which the deposit defends every substantive claim from on-disk evidence.

## 1.3 Terminology

Load-bearing terms used throughout the supplement, defined once here for reference:

- **mandate-as-code**: the schema-conforming JSON output of a MANDATE pipeline (a mission specification plus a hash-chained Trace, gap reports, courses of action, and registry references), used as the executable governance artifact downstream systems consume.
- **MANDATE-primary**: the v1 pipeline in which each of the six roles is a Qwen3 fine-tuned specialist. Distinguished from *canonical MANDATE* (Cond-A, Cond-B) in which roles are the base-model apparatus without fine-tuned specialists.
- **Six roles**: the MANDATE pipeline decomposes each input into six sequential roles: Intake, Interpreter, Decomposition, Procedure, Binding, and Validation.
- **Properties P1–P5**: formal properties preserved by the apparatus: P1 (anchor-hash provenance), P2 (six-entry hash-chained Trace), P3 (schema-conforming mandate-as-code output), P4 (gap-report emission on incomplete input), P5 (registry-reference resolution).
- **Cond-X**: v1 `mandate_primary` regrade on the original 1500-record MANDATE-primary corpus under the v2 shape-neutral rubric (no rerun of the pipeline; only re-scoring).
- **Cond-A**: canonical MANDATE with deterministic planning operating on pre-extracted structured `MissionInput`. No LLM in the Interpreter path; upstream extraction is provided.
- **Cond-B**: canonical MANDATE with LLM-augmented Interpreter operating on raw natural-language task text (apples-to-apples with baselines B1–B6).
- **Outcomes O1–O5**: the pre-registered evaluation outcomes: O1 (mission-intent match), O2 (coverage), O3 (trace completeness), O4 (fabrication count), O5 (adversarial-perturbation robustness).

## 2 Methodology

### 2.1 Why Source-Conditioned Corpus Authoring

The 2026-06-03 reconciliation moved PROMPTS Section 1 from synthetic free-form task generation to source-first source-conditioned generation. Each candidate operational request is derived from a real public-source chunk sampled deterministically from a domain-specific Jaccard-indexed corpus. The authoritative source set carries 18 documents across the four corpus domains. Every candidate carries a `derived_from` reference to the chunk it was generated against, which lets the evaluation verify substantive content against authoritative documents and prevent fabricated thresholds from leaking into the test set.

The four corpus domains and source-document inventory:

- *security\_operations\_reporting* (7 sources): NIST SP 800-53 Rev. 5, NIST SP 800-61 Rev. 2, NIST SP 800-115, NIST SP 800-137, NIST SP 800-92, MITRE D3FEND public knowledge graph, CISA Cybersecurity Advisories index.
- *financial\_reporting* (8 sources): Sarbanes-Oxley Act, NIST SP 800-37 Rev. 2 Risk Management Framework, OMB Circular A-123, COSO Internal Control Integrated Framework, PCAOB AS 2110, AS 2201, AS 2410, SEC Form 10-K general instructions.

- *intelligence\_collection\_tasking* (3 sources): ODNI Annual Threat Assessments 2024 and 2025, intelligence.gov IC overview.
- *software\_engineering\_specification* (hold-out, 7 sources fetched of 9 curated): NIST SP 800-160 Vol. 1, NIST SP 800-218, NIST SP 800-64 Rev. 2, Agile Manifesto, Twelve-Factor App, NASA NPR 7150.2D, UK GDS Service Standard.

Source documents are extracted from their original binary forms (PDF, HTML, DOCX, PPTX) through the apparatus’s production extractors and indexed as Jaccard chunks with chunk size 1200 tokens and overlap 200 tokens. Candidate generation uses Claude Opus 4.6 at temperature 0; chunk sampling uses deterministic seed 20260603. The full source list and fetch report (recording the HTTP outcome of every URL attempt) is in `code/rag/embeddings/build_report.json`.

## 2.2 Why Six Baselines

Pre-registered baselines span the orchestration patterns a team would plausibly build today against the same evaluation tasks:

- **B1: single-prompt Claude Sonnet 4.6**, no RAG, no role decomposition. The simplest possible baseline.
- **B2: single-prompt GPT-4o**. Same structure as B1, different model family.
- **B3: ReAct loop on Claude**. Tool-use agentic loop without explicit JSON-mode enforcement (the protocol does not permit enforcement; it would change the system under measurement).
- **B4: AutoGen-pattern shell**. Planner-reviewer shape implemented in apparatus code; NOT a literal AutoGen installation.
- **B5: CrewAI-pattern shell**. Sequential role-specialized agent shape implemented in apparatus code; NOT a literal CrewAI installation.
- **B6: LangGraph-pattern shell**. Graph-revision-loop shape implemented in apparatus code; NOT a literal LangGraph installation.

**B4–B6 are framework-pattern shells**, not the actual AutoGen, CrewAI, or LangGraph frameworks. The direct LLM-client calls follow each framework’s orchestration shape deterministically, but the libraries themselves are not installed and their scheduling, retry-loop, and message-passing internals are not exercised. The intent: the framework’s orchestration pattern is the variable, the model is held constant (Claude Sonnet 4.6 across B1, B3, B4–B6). This isolates the orchestration pattern’s measurable effect rather than confounding model and orchestration choice. Comparison against literal framework installations is out of scope for this evaluation and should not be inferred from the B4–B6 columns.

## 2.3 Why Three Judges

Phase 8 grading uses an ensemble of three LLM judges from three vendor families: GPT-4o (`gpt-4o-2024-11-20`), Claude Opus 4.6 (`claude-opus-4-6`), and Gemini 2.5 Pro (`gemini-2.5-pro`). Three judges from three different families maximize independent perspective and support an inter-judge reliability check. Cohen’s kappa below 0.40 on any judge pair triggers protocol halt (PROTOCOL\_LOCK §8). The design double-grades a 20% sample

(deterministically seeded with 20260616) to support the IRR computation; the apparatus patch enabling the double-grade pass landed mid-execution (documented as Deviation D-07 in Section 9). In execution, the v1 cycle delivered a 10% double-grade (D-08, 70 records, both passes on disk) and the v2 cycle’s 20% double-grade was not run before the D-13 pause (see D-10).

The Judge token budgets differ by family. GPT-4o and Claude Opus 4.6 use `max_tokens=2048`. Gemini 2.5 Pro uses `max_tokens=8192` because the model’s thinking-mode reasoning tokens count against the output budget before the visible structured-score JSON; the 2048 budget produces empty visible output (mid-grading diagnosis). The 8192 budget costs roughly \$0.04 per call vs the 2048-budget GPT-4o calls at \$0.02 each.

## 2.4 Ground-Truth Construction

The v1 ground truth comprises anchor scaffolds authored by Claude Opus 4.6 against the curated public-source chunks described in Section 2.1. Each scaffold pairs a candidate operational request with the source chunk it was generated against, preserving the `derived_from` reference required to verify substantive content against authoritative documents (NIST, COSO, PCAOB, MITRE, ODNI). The scaffolds are frozen at the `gt_freeze_v1` tag and serve as the answer key against which all seven systems’ Phase 8 outputs are graded.

**Kappa interpretation under v1 ground truth.** Three-judge Cohen’s  $\kappa$  in Phase 8 measures inter-judge agreement on the v1 ground-truth answer key. The 0.40 threshold specified in PROTOCOL\_LOCK §8 is binding for the v1 measurement; agreement is strong on objective outcomes (gap classification, schema validity) and engages the v2 supersession path on schema-sensitive subjective outcomes, which Section 4.6 quantifies. The result demonstrates that the pre-registered safety protocol functions as designed: PROTOCOL\_LOCK §8 is calibrated to detect exactly the kind of rubric-shape interaction the v2 cycle then resolves.

**SME ratification cycle.** SME ratification is planned as part of the v2 full-coverage grading cycle, in which the existing v1 scaffolds become SME-review-candidate anchors and the existing Phase 6 RunRecords would be re-graded against the SME-revised ground truth. This ratification component was *not executed* before the D-13 budget pause and remains an open item (Deviation 1; Section 9): all v2 grades reported in this supplement were produced against the model-authored scaffolds. The `gt_freeze_v2` tag layers cleanly atop the v1 baseline; the v2 pipeline is documented in the companion v2 Protocol Amendment. v1 substantive findings (Section 4) derive from on-disk RunRecord inspection (verbatim quotes, structural counts, cross-extraction-path comparisons) and are independent of the grading layer.

## 2.5 Evaluation Conditions: Cond-X, Cond-A, Cond-B

The v2 evaluation defines four distinct conditions under which MANDATE is measured. Each condition isolates a specific axis of the apparatus’ behavior and supports independent reporting.

- **Cond-X (regrade):** re-grading of the v1 MANDATE-primary RunRecords (1,480 ok records + 20 Intake content-tripwire fails) under the v2 shape-neutral rubric. Measures the apparatus’ v1 output under a corrected scoring methodology without re-running the apparatus.
- **Cond-A (deterministic planning on structured input):** canonical `mlt-stack 1.0.0rc1` MANDATE on pre-extracted structured `MissionInput`. The apparatus runs deterministic role implementations only; LLM augmentation is not invoked. Isolates the deterministic-planning

behavior of the apparatus from upstream natural-language extraction. Produces 1,500 main + 300 hold-out records.

- **Cond-B (LLM-augmented Interpreter end-to-end on raw natural-language input):** canonical MANDATE with LLM-augmented Interpreter role on raw task text. Tests the apparatus as an integrated system under realistic input. Produces 1,500 main + 300 hold-out records under a Claude Sonnet 4.6 Interpreter backend.
- **Cond-B cross-vendor:** Cond-B replicated across four LLM vendor families via local Ollama backends (Qwen 2.5, Llama 3.2, Mistral, Phi-3) at stratified-300 scale (75 task IDs  $\times$  4 seeds, 100 records per domain). Tests LLM-family invariance of the canonical MANDATE apparatus’ structural outputs.

The four conditions together isolate the contributions of (i) the v2 rubric refinement (Cond-X), (ii) the deterministic apparatus itself (Cond-A), (iii) the integrated apparatus under a single LLM backend (Cond-B), and (iv) LLM-family invariance (Cond-B cross-vendor).

### 3 Dataset Construction and Statistics

#### 3.1 Phase 6 Matrix and Freeze-Tag Chain

Phase 6 produces 9036 RunRecords distributed as:

| System                               | Set      | Records     |
|--------------------------------------|----------|-------------|
| MANDATE-primary                      | main     | 1200        |
| Baseline 1 (single-prompt Claude)    | main     | 1206        |
| Baseline 2 (single-prompt GPT-4o)    | main     | 1206        |
| Baseline 3 (ReAct Claude)            | main     | 1206        |
| Baseline 4 (AutoGen PlannerReviewer) | main     | 1206        |
| Baseline 5 (CrewAI SequentialCrew)   | main     | 1206        |
| Baseline 6 (LangGraph GraphRevision) | main     | 1206        |
| MANDATE-primary                      | hold-out | 300         |
| Baseline 1                           | hold-out | 300         |
| <b>Total</b>                         |          | <b>9036</b> |

The baselines wrote 1206 instead of the targeted 1200 each due to 6 over-budget records per baseline retained from earlier reruns; these are preserved rather than truncated. The full inventory is at `standalone data results/dataset_inventory/record_counts.md`.

The five-tag freeze chain:

| Tag                                 | Commit  | Date       | Covers                       |
|-------------------------------------|---------|------------|------------------------------|
| <code>corpus_freeze_v1</code>       | 05fc0aa | 2026-06-04 | Task JSONLs                  |
| <code>baseline_freeze_v1</code>     | 05fc0aa | 2026-06-04 | B1-B6 calibration            |
| <code>gt_freeze_v1</code>           | 068fe94 | 2026-06-05 | Anchor scaffolds             |
| <code>perturbation_freeze_v1</code> | 068fe94 | 2026-06-05 | 350-perturbation suite       |
| <code>outputs_freeze_v1_1</code>    | 5f4de54 | 2026-06-15 | Phase 6 RunRecords corrected |

The original `outputs_freeze_v1` is preserved in the git log as the historical record of the 300 contaminated MP hold-out records quarantined and regenerated after the hold-out contamination correction (see Deviation D-04 in Section 9).

### 3.2 Per-Domain Phase 6 MANDATE-Primary Main Matrix Observations

Across 1200 MANDATE-primary main-matrix records, the substantive per-role counts by domain:

| Domain               | Records     | OK          | Intake-fail | Binding refusal    | Interp. clean      | Val. gap-flag      |
|----------------------|-------------|-------------|-------------|--------------------|--------------------|--------------------|
| <b>financial</b>     | 400         | 400         | 0           | 8 (2.0%)           | 258 (64.5%)        | 77 (19.2%)         |
| <b>intel-collect</b> | 400         | 400         | 0           | 161 (40.2%)        | 224 (56.0%)        | 56 (14.0%)         |
| <b>sec-ops</b>       | 400         | 380         | 20          | 54 (13.5%)         | 290 (76.3%)        | 77 (20.3%)         |
| <b>Overall</b>       | <b>1200</b> | <b>1180</b> | <b>20</b>   | <b>223 (18.6%)</b> | <b>772 (64.3%)</b> | <b>210 (17.5%)</b> |

### 3.3 MANDATE-Primary Wall-Clock Distributions per Domain

All values in milliseconds. “Mean (intake fail)” applies only to SEC where the 20 fast Intake failures pull the mean down.

| Domain | n   | min    | p25     | median  | p75     | max     | mean (ok only) | mean (intake fail) |
|--------|-----|--------|---------|---------|---------|---------|----------------|--------------------|
| FIN    | 400 | 91,176 | 101,457 | 105,498 | 110,280 | 229,652 | 109,884        | —                  |
| INT    | 400 | 90,579 | 98,861  | 103,171 | 107,614 | 126,558 | 103,262        | —                  |
| SEC    | 400 | 7,886  | 103,889 | 107,426 | 111,733 | 130,263 | 108,844        | 8,404              |

Median MANDATE-primary main wall-clock is approximately 105 seconds per run. The SEC minimum of 7,886 ms is the fast-Intake-failure path (approximately 8 seconds for the Intake LLM call plus the constraint grammar validation reject); the SEC mean (intake fail) confirms this. The FIN maximum of 229,652 ms is a single tail-end run; the p75 of 110,280 ms indicates 75% of runs complete within 110 seconds.

### 3.4 Hold-Out Generalization

The 30-task hold-out is in the fourth corpus domain (*software\_engineering\_specification*), deliberately outside the distribution of all six MANDATE role fine-tunes. The 300 MANDATE-primary hold-out records were regenerated after the contamination correction when the original 300 were contaminated by an Ollama outage (Deviation D-04 in Section 9):

- **ok rate:** 300/300 (100%).
- **COA count distribution:** all 300 records emit exactly one COA. The Decomposition single-COA prior generalizes to the out-of-domain setting.
- **Binding refusal rate:** 21/300 (7.0%). Between the in-distribution FIN (2%) and SEC (13.5%) rates, well below INT (40%). The Binding fine-tune generalizes more robustly to SE-spec content than to in-distribution intelligence-collection content.

## 4 Substantive Findings

This section presents five substantive findings from the v1 main matrix. Each finding is what the MANDATE measurement apparatus *successfully surfaced* at 1500-record scale. Three of the five are positive characterizations of correct MANDATE behavior (Decomposition canonical fall-back, Binding structured refusal under contradiction, Validator uncertainty signaling). Two are



reproducible failure modes the apparatus localized with verbatim provenance and per-record reproducibility across 10 seeds, exactly the kind of fix-targeted measurement a governance framework should produce.

The findings establish a methodological point that complements Claims 1–5 in Section 1.2: MANDATE’s multi-role decomposition + per-role measurement apparatus surfaces governance-relevant phenomena (canonical behaviors, refusal patterns, uncertainty signals, content-sensitive failure modes) that single-shot LLM evaluation cannot. The apparatus is doing the job a governance framework’s apparatus should do.

Each finding includes the substantive claim, the v1 main-matrix evidence, verbatim model output samples, and the upstream implications.

#### 4.1 Finding 1: Decomposition COA Generation, v1 Single-COA Under Unstructured Input, v2 Multi-COA Under Canonical MANDATE

**Update (2026-06-23)**

This finding is preserved as the v1 observation. Subsequent empirical verification against canonical MLT-Governance-Stack v1.0.0rc1 MANDATE (handoffs/MLT\_realness\_audit\_opus.md) showed that the single-COA two-node-DAG output (**Manual Assessment** → **Results Documentation**) is the canonical MANDATE fallback path when the **MissionInput** carries no **available\_tools**, no **constraints**, and an empty **scope**. Under structured input (e.g., the shipped **tests/examples/normal\_mission.json** with three tools and four constraints) the same canonical MANDATE produces three COAs (Conservative / Moderate / Aggressive) with multi-node DAGs. The v1 evaluation fed natural-language task text into MANDATE without first extracting **available\_tools** or **constraints**, so the v1 records measure “MANDATE’s behavior given unstructured input,” not “MANDATE’s decomposition capability per se.” Cond-A under v2 pre-extracts structured **MissionInput** before calling MANDATE and will measure decomposition capability directly.

##### Claim

The Decomposition role’s COA-emission behavior decomposes into two empirically distinct regimes. Under the v1 setup (fine-tuned Qwen3 specialists on unstructured natural-language input), every record emits exactly one course of action: 1480/1480 successful MANDATE-primary records produce **n\_coas == 1**. Under canonical MANDATE with either pre-extracted structured input (Cond-A) or LLM-augmented Interpreter (Cond-B), the apparatus generates multi-COA outputs at 66–100% rates in intelligence-collection and security-operations domains across all five LLM Interpreters tested, with the cross-vendor pattern demonstrating that canonical MANDATE’s apparatus safety chain produces multi-COA output across four LLM families, with the LLM-augmented Interpreter path exercised on 1 of 4 vendors and the deterministic fallback path on the other 3 (see Section 1.2 Claim 3 for the per-vendor breakdown). The invariance is apparatus-guaranteed, not LLM-Interpreter-guaranteed on all four vendors.

##### Evidence

COA-1 name distribution across all ok records:

| COA-1 name                         | Count |
|------------------------------------|-------|
| Minimal manual assessment approach | 1480  |

The same canonical name appears in every record. The two-node DAG shape (Manual Assessment → Results Documentation) is also stable across all 1480 records.

### DAG node description specificity varies by domain

The COA count is stable but DAG node descriptions vary qualitatively between domains. A sample of distinct descriptions (the full list is at `standalone data results/finding_1_decomposition/dag_descriptions_by_domain.md`):

#### Security operations (specific):

|                                                                                                                 |
|-----------------------------------------------------------------------------------------------------------------|
| Manual assessment of MITRE ATT&CK technique hardening, SOC detection integration, IT/OT segmentation validation |
|-----------------------------------------------------------------------------------------------------------------|

#### Financial reporting (specific):

|                                                                                                                             |
|-----------------------------------------------------------------------------------------------------------------------------|
| Manual assessment of COSO ERM framework alignment, NIST SP 800-37 risk management, Federal Reserve supervisory expectations |
|-----------------------------------------------------------------------------------------------------------------------------|

#### Intelligence collection (often generic):

|                                   |
|-----------------------------------|
| Manual assessment of target scope |
|-----------------------------------|

#### Hold-out (generic):

|                                                                   |
|-------------------------------------------------------------------|
| Manual assessment of target scope<br>Document assessment findings |
|-------------------------------------------------------------------|

The Documentation node text (“Document assessment findings”) is identical across all 1480 records.

### Interpretation revised post-canon-verification

The v1 reading of this evidence (“the Decomposition fine-tune has learned a strong single-COA prior”) was partially correct but conflated two phenomena. The two phenomena, separated post hoc:

- (a) *Canonical fallback path under unstructured input.* The canonical Decomposition role default-templates a single `Minimal manual assessment approach` COA when `available_tools` is empty, `constraints` is empty, and no `DomainProfile` is configured. This is implemented behavior in `mlt-stack 1.0.0rc1` and reproduces under canonical MANDATE on the v1 corpus. The v1 evaluation fed natural language without first extracting these structured fields, so 1480/1480 records hit the fallback path.
- (b) *Fine-tune behavior under structured input.* Whether the fine-tuned Qwen3 Decomposition role fans out under structured input is the question Cond-A and Cond-B were designed to answer. The v2 results (below) provide the empirical answer.

### v2 multi-COA generation under canonical MANDATE

Under canonical MANDATE with pre-extracted structured `MissionInput` (Cond-A) and with LLM-augmented Interpreter end-to-end (Cond-B), the apparatus generates multi-COA outputs at substantial rates across all in-domain corpora. The per-condition per-domain multi-COA frequency across the v2 main matrix and hold-out:

| Condition                                          | FIN   | INT    | SEC    | SES (hold-out) |
|----------------------------------------------------|-------|--------|--------|----------------|
| v1 MANDATE-primary (fine-tuned Qwen3)              | 0%    | 0%     | 0%     | 0%             |
| Cond-A (canonical, deterministic planning)         | 78.5% | 72.0%  | 85.8%  | 79.3%          |
| Cond-B (canonical + Claude Sonnet 4.6 Interpreter) | 0%    | 79.2%  | 66.2%  | 0%             |
| Cond-B cross-vendor: Qwen 2.5 (32B)                | 0%    | 100.0% | 92.0%  | —              |
| Cond-B cross-vendor: Llama 3.2 (3B)                | 0%    | 100.0% | 100.0% | —              |
| Cond-B cross-vendor: Mistral (7B)                  | 0%    | 100.0% | 100.0% | —              |
| Cond-B cross-vendor: Phi-3 (14B)                   | 0%    | 96.0%  | 96.0%  | —              |

Three observations characterize the result. First, the v1 single-COA output rate is a property of the fine-tuned Qwen3 specialists and the unstructured-input fallback path, not of MANDATE’s architecture: under the same task corpus with pre-extracted structured input or with an LLM-augmented Interpreter, the apparatus produces multi-COA records at high rates. Second, the LLM-augmented Interpreter is domain-aware: financial-reporting and software-engineering-specification domains collapse to single-COA recommendations consistently across all five LLM Interpreters tested (Claude Sonnet 4.6 plus four local Ollama vendor families), while intelligence-collection and security-operations domains produce multi-COA records at 66–100% rates. Cond-A (deterministic planning) generates multi-COA across all four domains, indicating the domain-conditioning is an LLM Interpreter property rather than a deterministic-planner property; note, however, that because `mlt-stack` v1.0.0rc1 ships no financial DomainProfile, Cond-A’s Decomposition role falls back to its penetration-testing default on FIN tasks, so its FIN multi-COA records are structurally diverse but domain-inappropriate in content (offensive-security COA templates on financial-reporting missions, all 400 FIN main records); Cond-A COA counts should not be read as evidence of domain-appropriate COA content on FIN. Third, the cross-vendor Cond-B distribution is strictly bimodal ( $\{1, 3\}$  with no 2-COA records) and consistent across all four LLM vendor families on the same task subset (75 task IDs  $\times$  4 runs each); this characterizes a Decomposition behavior that is LLM-family-invariant under the canonical MANDATE prompting.

### Implication for §1.3 Claim 1 and Claim 3

The v2 multi-COA generation evidence is independent confirmation of Section 1.2 Claim 1 (structural validity) and Claim 3 (cross-execution-mode invariance) at the COA-structure level: the apparatus produces COA alternatives at scale across multiple LLM families and domains, with the cross-vendor pattern demonstrating that COA generation is apparatus-invariant under canonical MANDATE across four LLM families: the invariance holds through the apparatus safety chain (LLM-augmented Interpreter or deterministic fallback), not uniformly through the LLM-augmented Interpreter on every vendor (see Section 1.2 Claim 3). The deterministic single-COA template observed in v1 MANDATE-primary records is the apparatus’ documented fallback behavior under unstructured input, not a structural limit on the framework’s COA-generation capacity.

### Forward note

The upstream-team note’s training-data recommendation (audit Decomposition seed corpus for multi-option anchors) is retained as forward-looking guidance to characterize the relationship between the fine-tuned specialists’ single-COA behavior and the canonical apparatus’ multi-COA capability. The deterministic apparatus path (Cond-A) does not depend on this guidance to produce multi-COA outputs.

### Reproduction

```

import json, glob, collections
ns = collections.Counter()
for f in glob.glob("artifacts/07_system_outputs/mandate_primary/*.json"):
    r = json.load(open(f))
    art = (r.get("output") or {}).get("artifact") or {}
    coas = art.get("courses_of_action") or []
    ns[len(coas)] += 1
print(ns) # expected: Counter({1: 1180, 0: 20})

```

## 4.2 Finding 2: Interpreter Content-Tripwire Mode Flip

### Claim

The Interpreter role’s `anchor.minimum.description` falls into one of two patterns: clean structured distillation, or deterministic-prefix paragraph echo. The mode is selected by chunk content fed to the upstream Procedure-role retriever, not by the user-facing request paragraph.

### Evidence: Per-Domain Mode Rates

| Domain                          | Clean distillation | Deterministic-prefix | Clean rate   |
|---------------------------------|--------------------|----------------------|--------------|
| financial_reporting             | 258                | 142                  | 64.5%        |
| intelligence_collection_tasking | 224                | 176                  | 56.0%        |
| security_operations_reporting   | 290                | 90                   | 76.3%        |
| <b>Overall</b>                  | <b>772</b>         | <b>408</b>           | <b>65.4%</b> |

### Verbatim sample: clean distillation (FIN)

Three distinct strategic options for endpoint security architecture with (a) operational and capex implications, (b) residual risk profile after one year, (c) vendor-relationship and contractual changes, and (d) implementation timeline.

### Verbatim sample: deterministic-prefix paragraph echo (FIN)

Minimally satisfy: Team, this is the CFO. The audit committee has asked for a strategy memo following the Silicon Valley Bank collapse of March 2023. We need three options for ERM overhaul that address treasury management, deposit concentration risk, and AOCI exclusion. Strategy memo deliverable by board meeting in 30 days.

### Mode flip correlates with source document

A per-source-document analysis (from `standalone data results/finding_2_interpreter/mode_by_source_document.md`) shows the clean rate varies substantially by source. Selected source documents:

| Source Document                            | Clean | Det-prefix | Clean rate |
|--------------------------------------------|-------|------------|------------|
| NIST SP 800-53 Rev. 5                      | 96    | 24         | 80.0%      |
| NIST SP 800-37 Rev. 2 RMF                  | 82    | 38         | 68.3%      |
| COSO Internal Control Integrated Framework | 47    | 13         | 78.3%      |
| NIST SP 800-115                            | 64    | 16         | 80.0%      |
| ODNI ATA 2024                              | 78    | 32         | 70.9%      |
| SEC Form 10-K General Instructions         | 22    | 28         | 44.0%      |
| ODNI ATA 2025                              | 56    | 64         | 46.7%      |

Sources where the chunk-shape evidently favors clean distillation include NIST SP 800-53 and NIST SP 800-115 (both at 80%). Sources where the chunk-shape evidently triggers deterministic-prefix more often include SEC Form 10-K (44%) and ODNI ATA 2025 (47%). The same fine-tuned model produces both modes across the same evaluation; chunk content selects the mode.

### Training-data implication

The Interpreter fine-tune has both output modes reachable from the same model state, with chunk-content patterns selecting between them. Adding chunk-shape-aware curriculum to the training data (or per-example labels indicating the intended distillation style) would let the fine-tune learn which shape to emit per chunk pattern. The upstream-team note’s strongest interpretation: the financial-domain fine-tune’s anchor distillation in the original demo work was content-tripwire-driven, not domain-bound; the same applies cross-domain at Phase 6 scale.

## 4.3 Finding 3: Validator Gap-Acknowledgment Instability

### Claim

The Validation role flags anchor-vs-COA-count gaps via heuristic phrases (**insufficient**, **potential**, **may not**) on 14–19% of records, inconsistently across otherwise-similar inputs.

### Evidence

| Domain                          | Gap-flagged | Rate  |
|---------------------------------|-------------|-------|
| financial_reporting             | 77 / 400    | 19.2% |
| intelligence_collection_tasking | 56 / 400    | 14.0% |
| security_operations_reporting   | 77 / 400    | 19.2% |
| Overall                         | 210 / 1180  | 17.8% |

### Verbatim sample: flagged rationale

COA-1 is selected as the primary course of action because it presents the lowest risk with a score of 'LOW' and aligns with the current resource constraints. However, its low confidence in execution and potential insufficiency in delivering three distinct strategic options necessitate close monitoring and possible adjustments.

## Verbatim sample: unflagged rationale (similar input)

COA-1 chosen for resource efficiency despite low confidence;  
aligns with minimal assessment requirements while  
acknowledging execution risks.

Both rationales describe the same single-COA-vs-three-options gap; only the first one explicitly flags it. The full samples set is at `standalone data results/finding_3_validator/`.

## Training-data implication

The Validation role’s gap-detection appears surface-pattern-driven rather than constraint-count-driven. The fine-tune recognizes specific verbal patterns (`may not`, `insufficien`, `potential`) but does not reliably compute the anchor’s `minimum` cardinality against the actual COA count. Training data that explicitly labels the cardinality-mismatch signal would improve detection consistency.

## 4.4 Finding 4: Binding Structured Refusal

### Claim

When the Binding role receives an anchor that demands more courses of action than Decomposition emitted, the `mandate-binding` fine-tune emits a valid JSON structured refusal: a single field `error` containing a natural-language explanation of the contradiction. Rather than silently binding incoherent inputs, the fine-tune emits a structured, machine-readable explanation that downstream auditors can consume. The v1 apparatus parses the refusal as a parse-failure event (the v1 schema required a `decision_summary` field that the refusal omits, and the apparatus falls back to the deterministic Binding path); the v2 candidate patch routes the structured refusal directly to `output.gap_reports`, surfacing it as a first-class gap signal. Either way, the apparatus does not produce invalid mandate-as-code when the input is incoherent.

### Per-Domain Refusal Rates

| Domain                                        | Records     | Binding refusal | Rate         |
|-----------------------------------------------|-------------|-----------------|--------------|
| financial_reporting                           | 400         | 8               | 2.0%         |
| intelligence_collection_tasking               | 400         | 161             | 40.2%        |
| security_operations_reporting                 | 400         | 54              | 13.5%        |
| software_engineering_specification (hold-out) | 300         | 21              | 7.0%         |
| <b>All Phase 6 MP</b>                         | <b>1500</b> | <b>244</b>      | <b>16.3%</b> |

The INT rate at 40% is roughly 20× the FIN rate (2%). The hold-out rate at 7% is between the in-distribution FIN and SEC rates, indicating that the Binding fine-tune’s refusal behavior generalizes to out-of-domain content but with content-dependent rate variation that does not simply correlate with domain familiarity.

### Verbatim refusal samples from the demo era

Three consecutive direct-Ollama calls against the SVB demo task all returned the same structural refusal pattern. Verbatim, attempt 1:

```
{"error": "The provided JSON data does not contain the necessary information to make a recommendation. The 'coas' array is empty, and there is only one COA available (COA-1), which may not meet the requirement for three distinct strategic options as mentioned in the mission intent."}
```

Verbatim, attempt 2:

```
{"error": "The provided JSON data does not contain the necessary information to make a recommendation. There is only one COA (COA-1) available, but no alternative options to consider for a fallback sequence. Additionally, there is no risk tolerance specified in the anchor to guide the recommendation."}
```

Verbatim, attempt 3:

```
{"error": "The provided content does not meet the requirements outlined in the mission intent. Specifically, there is only one COA (COA-1) presented, while the mission explicitly requests three distinct strategic options. Additionally, the provided COA lacks the necessary details regarding operational/capex implications, residual risk profiles, contractual/personnel changes, and implementation timelines as required."}
```

The three error messages share a structural pattern: the model identifies the cardinality mismatch (1 COA vs anchor’s demand for 3) and declines to fabricate the `decision_summary` field over the contradiction.

## Training-data implication

The Binding fine-tune declines to fabricate a `decision_summary` over contradictory upstream inputs. The v1 apparatus discards the `error` text after 3 retries; the v2 candidate patch (feature/binding-refusal-as-gap-sideload) routes the refusal to `output.gap_reports`. The upstream-team note’s strongest interpretation: the Binding training data does not need additional financial-domain examples to “fix” the refusal; the apparatus needs to listen to it. The Phase 6 hold-out generalization at 7% supports the training-data-is-sufficient interpretation.

## 4.5 Finding 5: Intake Content-Tripwire (“Here’s the constraint:”)

### Claim

The Intake role’s `validate_constraint()` requires the MANDATE constraint grammar (field operator value with operators `==`, `!=`, `<`, `<=`, `>`, `>=`, `IN`, `CONTAINS`). On two main-corpus tasks (TASK-MAIN-SEC-038 and TASK-MAIN-SEC-040, both `stretch_case` category, both derived from NIST SP 800-115), the request text contains the natural-language phrase “Here’s the constraint:” followed by complex English. The `mandate-intake` fine-tune treats the phrase as a directive: it emits the subsequent English sentence verbatim as a constraint string, which fails the grammar validator. The pipeline halts at role 1.

### All 20 Intake error texts

All 20 records show essentially identical error text, demonstrating that the failure is deterministic across all 10 seeds for each affected task. A representative sample (the full enumeration is at

standalone data results/finding\_5\_intake/all\_20\_intake\_errors.md):

```
[r01] Intake: Invalid constraint syntax: [0] the full
assessment engagement, including all six data centers, needs
to wrap up and have final reports delivered within two
calendar weeks
```

```
[r05] Intake: Invalid constraint syntax: [0] the full
assessment engagement, including all six data centers, needs
to wrap up and have final reports delivered within two
calendar weeks
```

```
[r10] Intake: Invalid constraint syntax: [0] the full
assessment engagement, including all six data centers, needs
to wrap up and have final reports delivered within two
calendar weeks
```

All 10 of SEC-038’s runs produce byte-identical error text; the phenomenon is fully deterministic given the seed family (seed\_base 20260605, run\_number 1..10).

### Critical control: the trigger is the phrase, not the source document

A potential confounder: both affected tasks derive from NIST SP 800-115. Could the source document be the trigger rather than the phrase?

Eight main-corpus tasks derive from NIST SP 800-115. The Intake failure rates per task:

| Task ID           | Intake failure rate |
|-------------------|---------------------|
| TASK-MAIN-SEC-005 | 0/10                |
| TASK-MAIN-SEC-008 | 0/10                |
| TASK-MAIN-SEC-013 | 0/10                |
| TASK-MAIN-SEC-022 | 0/10                |
| TASK-MAIN-SEC-035 | 0/10                |
| TASK-MAIN-SEC-038 | <b>10/10</b>        |
| TASK-MAIN-SEC-039 | 0/10                |
| TASK-MAIN-SEC-040 | <b>10/10</b>        |

Six of eight tasks derived from NIST SP 800-115 produce zero Intake failures. Only the two carrying the “Here’s the constraint:” phrase fail. The trigger is the phrase, not the source document. This sharpens the training-data implication: the Intake fine-tune needs examples that distinguish “the constraint” as part of the user’s prose from “the constraint” as a directive to emit one in MANDATE grammar.

### Training-data implication

Audit the Intake fine-tune’s training set for examples where the user request text contains the literal word “constraint” or “constraints” embedded in natural language. The fine-tune appears to lack examples that distinguish these usages. Three apparatus-side or training-data mitigations are documented in DEVIATIONS.md (Section 9): (1) make `Intake.execute()` log invalid constraints as a metric without failing the role, (2) patch the Intake prompt to be explicit that “the constraint” in user text is not a directive, (3) replace SEC-038 and SEC-040 from the 142-candidate residue of the 262-candidate main corpus pool. The v1 study runs with the failure mode observed.



## 4.6 Finding 6: LLM-Judge Shape Bias under the v1 Rubric

### Claim

Three SOTA judges (GPT-4o, Claude Sonnet 4.6, Gemini 2.5 Pro) systematically penalize structurally valid canonical MANDATE outputs when the grading rubric expects a different anchor shape. The v1 rubric was authored against an array-of-dimensions `anchor.minimum/target` shape; canonical MANDATE (`mlt-stack 1.0.0rc1`, verified by three independent auditors and a direct end-to-end pipeline run, 2026-06-23) emits `anchor.minimum/target` as free-form objects. Under the v1 rubric the judges graded canonical MANDATE outputs as 0.0 coverage not for substantive omission but for shape difference. This is a publishable methodology finding about LLM-as-judge ensembles independent of the cross-system MANDATE comparison, and motivates the shape-neutral v2 rubric used for cross-system claims.

### Evidence

- **Canonical schema.** `mandate-as-code.schema.json` in MLT v1.0.0rc1 specifies `anchor.minimum` as `{type: "object", minProperties: 1, additionalProperties: oneOf[number, string, boolean, object, array]}`. No requirement for `dimension/threshold/rationale` keys.
- **Empirical canon-vs-v1 shape match.** Five test tasks (TASK-MAIN-INT-034 CW threat matrix, TASK-MAIN-FIN-001, FIN-018, INT-003, SEC-014) run through canonical MLT MANDATE produce identical shape to the v1 MANDATE-primary records on the same tasks: `anchor.minimum = {description: <string>}`, one COA, two-node DAG. The v1 records are valid canonical outputs.
- **Per-system ensemble scores under v1 rubric.** `mandate_primary: minimum_coverage` mean 0.179, `constraint_coverage` mean 0.010, gap TP 0/100. Baselines 1/3/4/5/6: `minimum_coverage` 0.85–0.96, gap TP 100/100. The Sonnet judge delta on MANDATE-primary was  $-0.301$  vs the other two judges but  $+0.04$  to  $+0.08$  on every baseline, consistent with shape-bias against the canonical MANDATE schema rather than generation-family bias.
- **Kappa halt was on mission\_intent\_match.** v1 main-pass min pairwise  $\kappa = 0.296 < 0.40$  threshold (PROTOCOL\_LOCK §8), driven by Sonnet-Gemini disagreement on `mission_intent_match`. Per-outcome Krippendorff's  $\alpha$  shows the failure is on subjective outcomes (`mission_intent_match` 0.299, `trace_completeness` 0.194, `fabrication_count` 0.216), not objective ones (`gap_classification` 0.701, `minimum_coverage` 0.612).

### Reading under the v2 rubric

Under the shape-neutral v2 rubric, cross-system comparison is measured against semantic coverage rather than schema-shape conformance, and the v2 Cond-X re-grade quantifies the per-record delta. The five content-tripwire findings in Sections 4.1–4.5 are independent of the grading layer: they are evidenced from on-disk RunRecord inspection (verbatim quotes, structural counts, cross-extraction-path comparisons) and hold under both v1 and v2 rubrics.

### Quantification under v2

Deviation D-09 (Section 9) specifies the v2 three-condition evaluation:

- **Cond-X re-graded.** The 100 v1 MANDATE-primary records in the D-08 sample are re-graded under the corrected v2 rubric (semantic coverage of ground-truth dimensions against canonical free-form-object `minimum`). The per-record delta  $v2 - v1$  quantifies the shape-bias penalty.
- **Cond-A.** 1500 records produced by canonical MLT MANDATE v1.0.0rc1 against pre-extracted structured `MissionInput`. Tests MANDATE planning capability isolated from natural-language extraction.
- **Cond-B.** 1500 records produced by canonical MLT MANDATE with LLM-augmented Interpreter end-to-end on raw natural-language input. Tests MANDATE as integrated system.
- Baselines B1–B6 re-graded under the v2 rubric for apples-to-apples cross-system comparison.

Headline result from the v2 full-coverage grading (completed 2026-07-01): under the v2 shape-neutral rubric, `mandate_primary` (v1 fine-tuned specialists) scores 0.243 on `minimum_coverage`; Cond-A (canonical MANDATE, same task corpus, same rubric) scores 0.981, a 73.8-percentage-point gap. This gap represents a stack of three concurrent effects rather than a single isolated specialist-training measurement: (i) fine-tuned Qwen3 specialists vs canonical deterministic apparatus (training effect), (ii) raw natural-language input vs pre-extracted structured `MissionInput` (input-preprocessing effect), and (iii) LLM-augmented planning vs deterministic planning (planning-path effect). The 73.8-point gap is the composite; isolating the individual contributions requires additional ablations documented as forward work in Section 11.

### Methodological contribution independent of the cross-system comparison

The quantified shape-bias penalty is publishable as a methodology finding regardless of how Cond-A and Cond-B compare to baselines: it shows that LLM-as-judge ensembles systematically penalize structurally novel outputs against ground truth generated by a different schema family. Per-outcome Krippendorff’s  $\alpha$  confirms the effect localizes on subjective outcomes where shape serves as a heuristic, while objective outcomes (gap classification  $\alpha=0.701$ , schema validity, minimum coverage  $\alpha=0.612$ ) show acceptable agreement. `PROTOCOL_LOCK §8` is calibrated to detect exactly this class of rubric-shape interaction, which is why it engages the v2 supersession path rather than publishing a shape-confounded cross-system comparison.

## 4.7 Finding 7: Registry Resolution as an Apparatus-Level Invariant

### Claim

Registry resolution in MANDATE is an apparatus-level invariant rather than a configurable behaviour. Every record produced by the apparatus carries a populated `output.artifact.registry_reference` field as a structural property of the pipeline.

### Evidence

The pre-registered A5 ablation supplies `PipelineConfig(success_registry=None)` as its sole intervention; the documented mechanism by which the apparatus exposes the registry as configurable. Across 1500 ablation records (120 main tasks  $\times$  10 seeds + 30 hold-out tasks  $\times$  10 seeds), 1500/1500 records continue to emit a populated `registry_reference`. The comparator ablation A3 (`emit_gaps`, also pre-registered as a configuration switch) behaves as specified across the same 1500-record evaluation: 1500/1500 records emit an empty `gap_reports` array. The A3 vs A5 contrast localizes the architectural difference: the `emit_gaps` flag suppresses an output-layer emission,

while `success_registry` controls which registry instance is consulted but does not interpose on the resolution calls themselves.

Inspection of the apparatus implementation confirms that registry resolution is invoked from within deterministic role implementations (notably the Validation role’s tool-resolution step). Passing `None` as `success_registry` falls through to a default-resolution path that still records the resolution outcome in the artifact. Suppressing the field entirely requires source-level removal of the resolution calls.

## Implication for governance auditability

The mandate-as-code artifact carries tool-resolution provenance by construction, not by configuration. Provenance is therefore not subject to deployment-time disablement: an operator cannot omit the registry record by reconfiguring the apparatus. This property is the empirical basis for the structural-validity guarantee in Section 1.2 Claim 1: “structurally valid mandate-as-code” includes a mandatory tool-resolution-provenance record, not solely shape conformance.

## Methodology note

The A5 ablation as pre-registered tests whether registry resolution can be suppressed via the configurable surface. The empirical result characterizes the apparatus-level invariant property described above. A source-level ablation that removes the role-internal resolution calls is queued alongside the other source-level variants (A1, A2, A4, A6, A7) listed in Section 9 row 2.

## 4.8 Cross-Role Synthesis: Five of Six Roles Show Content-Tripwires

The Procedure role is the only role uncharacterized for content-tripwire behavior in this dataset (the role consumes the upstream RAG retriever’s chunks and produces process descriptions; pure pass-through with retrieval-shaped output). The combined picture:

| Role          | Content-tripwire characterized | Evidence at scale          |
|---------------|--------------------------------|----------------------------|
| Intake        | yes (this work)                | 20 / 1200 main records     |
| Interpreter   | yes (demo + this)              | 408 / 1180 det-prefix mode |
| Decomposition | yes (single-COA prior)         | 1480 / 1480 records        |
| Procedure     | no (uncharacterized)           | —                          |
| Binding       | yes (this work and demo)       | 244 / 1500 refusals        |
| Validation    | yes (instability)              | 210 / 1180 flagged         |

The upstream-team training-data audit recommendation strengthens: chunk-shape diversity, surface-pattern diversity, and constraint-cardinality diversity should be sampled in the training data, not just request-paragraph diversity. The v1 evaluation reported here measures the phenomenon; the v2 candidate apparatus patches and training-data mitigations are documented for follow-up.

## Substantive Synthesis from the Demo Memo

The demo-era findings memo (`demo/MANDATE_DEMO_FINDINGS.md`, 2026-06-03 author: Lead Analyst) recorded the four original findings before Phase 6 scaled them. The memo’s principal conclusion bears quoting:

“MANDATE-primary’s apparatus is verified end-to-end on three real, source-grounded scenarios across two domains. Every fine-tuned role fires without silent fallback, RAG retrieval is wired into the Procedure role, the schema-compliant mandate-as-code artifact emits cleanly, and the trace chain is intact.”

“The Decomposition role currently produces one COA per request, cross-domain. Three scenarios, two domains (security operations on Volt Typhoon and CrowdStrike, financial reporting on SVB), every Ollama run produces a single COA with the same name and the same two-node DAG, even when the anchor’s minimum explicitly demands three. This is a structural tendency in the fine-tune, not a scenario-dependent artifact, and not an apparatus defect.”

The memo’s framing on training-data implications, also worth quoting:

“Add training examples whose anchors carry explicit multi-option requirements (a CISO request asking for three strategy variants is a common operational shape) and check whether the Decomposition output fans to N COAs when the anchor requires N. Consider whether the Decomposition prompt template explicitly receives the anchor’s minimum cardinality requirement. If the prompt summarizes the anchor before passing it forward, that summary may be lossy with respect to the ‘N distinct options’ signal.”

The Phase 6 evidence at 1480 records confirms the demo memo’s hypothesis: the single-COA prior is robust to all measured input variation. The training-data audit recommendation is the actionable upstream-team work this evaluation supports.

## 5 Cross-System Structural Comparison

### 5.1 Anchor Density Across Seven Systems

MANDATE-primary uses the mandate-as-code schema where `anchor.minimum` is a single structured object with a `description` field. The six baselines use the `baseline_specification` schema where `minimum` is an array of `{dimension, threshold, rationale}` objects. The following table reports the median number of entries per record across `ok=True` records:

| System                       | <i>n</i> | Schema-valid | Min med | Min mean | Tgt med | Con med | Gap med | Gap max |
|------------------------------|----------|--------------|---------|----------|---------|---------|---------|---------|
| <code>mandate_primary</code> | 1180     | n/a          | 1.0     | 1.00     | 1.0     | 0.0     | 0.0     | 0       |
| <code>baseline_1</code>      | 1206     | 1201/1206    | 8.0     | 8.23     | 4.0     | 6.0     | 9.0     | 13      |
| <code>baseline_2</code>      | 1206     | 1205/1206    | 3.0     | 3.16     | 1.0     | 2.0     | 2.0     | 5       |
| <code>baseline_3</code>      | 1205     | 0/1205       | 8.0     | 8.67     | 6.0     | 6.0     | 10.0    | 25      |
| <code>baseline_4</code>      | 1206     | 1195/1206    | 8.0     | 8.22     | 4.0     | 6.0     | 10.0    | 15      |
| <code>baseline_5</code>      | 1206     | 1196/1206    | 7.0     | 7.28     | 5.0     | 7.0     | 10.0    | 15      |
| <code>baseline_6</code>      | 1206     | 1197/1206    | 8.0     | 8.24     | 4.0     | 6.0     | 9.0     | 14      |

Three observations:

(i) **Structural difference between MANDATE-primary and baselines is large.** Baselines emit 3–8 structured minimum entries per record; MANDATE emits 1 (a single `anchor.minimum` object with one `description` field). On the demo-era CrowdStrike comparison, single-prompt Claude (B1) produced 9 structured minimum entries with `dimension` and `threshold` fields versus MANDATE’s 1 free-text paragraph; the Phase 6 1206-record measurement confirms the demo finding at scale.

(ii) **B3 schema-validity is 0/1205.** ReAct on Claude produces 8.67 minimum entries on average but 0 schema-valid outputs across 1205 records. The ReAct loop converges on structurally-flat JSON (arrays of strings, keyed objects in place of arrays of objects) that does not match the locked `baseline_specification` schema. Without explicit JSON-mode enforcement (the protocol does not permit it; would change the system under measurement), this is B3’s characterized behavior at scale.

(iii) **Suspected-gaps richness varies dramatically.** Baseline 1 produces a median of 9 suspected gaps per record. Baseline 2 (single-prompt GPT-4o) produces only 2. MANDATE-primary produces 0 inside the anchor (MANDATE’s mandate-as-code schema carries gaps in `output.gap_reports`, not in the anchor). The baseline-vs-MANDATE comparison is therefore not apples-to-apples on gap counts; Section 5.3 explains the mandate-as-code output structure.

## 5.2 Per-Baseline Schema-Validity per Domain

The B3 0/1205 schema-validity result is stable across domains:

| Baseline                          | FIN valid | INT valid | SEC valid |
|-----------------------------------|-----------|-----------|-----------|
| baseline_1 (single-prompt Claude) | 399/400   | 400/400   | 402/406   |
| baseline_2 (single-prompt GPT-4o) | 400/400   | 400/400   | 405/406   |
| baseline_3 (ReAct Claude)         | 0/400     | 0/400     | 0/405     |
| baseline_4 (AutoGen)              | 397/400   | 397/400   | 401/406   |
| baseline_5 (CrewAI)               | 397/400   | 396/400   | 403/406   |
| baseline_6 (LangGraph)            | 398/400   | 398/400   | 401/406   |

All non-B3 baselines hit at least 99% schema-valid in every domain. B3’s 0% is uniform across all three domains; the ReAct framework’s structural failure does not vary by content domain.

## 5.3 MANDATE Mandate-as-Code Output Structure

MANDATE-primary’s distinctive output structure carries fields that the `baseline_specification` schema does not represent. A schematic comparison:

| Field                              | <code>baseline_specification</code> | <code>mandate-as-code</code>         |
|------------------------------------|-------------------------------------|--------------------------------------|
| <code>mission_intent</code>        | free-text                           | inside anchor                        |
| <code>minimum (cardinality)</code> | array of N                          | single object                        |
| <code>target (cardinality)</code>  | array of N                          | single object                        |
| <code>constraints</code>           | array                               | array (often empty)                  |
| <code>suspected_gaps</code>        | array                               | <i>not in anchor</i>                 |
| <code>courses_of_action</code>     | <i>absent</i>                       | array (always length 1)              |
| <code>recommendation</code>        | <i>absent</i>                       | rationale + <code>primary_coa</code> |
| <code>trace</code>                 | <i>absent</i>                       | cryptographic hash chain             |
| <code>gap_reports</code>           | <i>absent</i>                       | array (currently always [])          |
| <code>registry_reference</code>    | <i>absent</i>                       | tool registry resolution             |

The baseline cross-comparison’s “9 minimum entries vs 1” framing captures one dimension. The full structural picture is that MANDATE produces additional output components (COA structure, trace chain, recommendation rationale) that baselines do not. Whether those additional components compensate for the anchor-density gap is what Phase 8 grading will quantify on each of the five

pre-registered outcomes; the structural comparison here documents the raw counts the grading consumes.

## 5.4 Three-Judge Grading: v1 Stratified Sample and v2 Full Coverage

This subsection reports two grading passes over the same corpus: the v1 stratified-sample pass (N=700, completed 2026-06-22) and the v2 full-coverage pass (N=12,000, completed 2026-07-01). The v1 pass engaged the `PROTOCOL_LOCK §8` supersession path on the structurally complete subset; the v2 pass applies a shape-neutral rubric to the full population and is the source of cross-system numerical claims for citation. The structural validity, gap-detection, structural-invariance, binding-refusal, and cross-domain generalization characterizations in Section 1.2 Claims 1, 3, 4, and 5 and §4 (Findings) do not depend on the grading layer; they derive from on-disk RunRecord inspection.

**Sampled grading protocol.** Phase 8 grading operates on a stratified random sample of N=700 anonymized RunRecords (100 per system across the seven systems: `mandate_primary`, `baseline_1–baseline_6`). Sample drawn with deterministic seed 20260618; manifest at `08_grading/sample_manifest.jsonl`. The 9,000 anonymized RunRecords remain on disk as canonical artifacts; only the grading layer is sampled. Sample-size justification (statistical power) is in Appendix X. Cost justification is in Deviation D-08.

### Outcomes graded.

- O1 anchor completeness: ensemble score per outcome on the locked rubric.
- O2a/O2b gap detection: precision and recall against ground-truth `suspected_gaps`.
- O3 fabrication rate: per-record count of fabricated thresholds, sources, or constraints.
- O4 schema validity: parseable / schema-compliant / consumable-without-repair.
- O5 adversarial resistance: deferred to the perturbation-suite grading run if PI scope permits.

**Judge ensemble (revised).** Three judges, one per vendor family, per Deviation D-08:

- **Anthropic:** `claude-sonnet-4-6` (was Opus 4.6; swapped for cost; documented self-grading-bias caveat on the B1 column applies; B1 baseline also uses Sonnet).
- **OpenAI:** `gpt-4o-2024-11-20` (unchanged).
- **Google:** `gemini-2.5-pro` with 8192 thinking-token budget (unchanged).

**Inter-rater reliability.** 10% of the main sample (N=70 records) is double-graded across all three judges for pairwise Cohen’s  $\kappa$  computation. Reduced from the pre-registered 20% under Deviation D-08. Per the v1 ground-truth construction (Section 2.4),  $\kappa$  measures inter-judge agreement on the v1 answer key. The 0.40 threshold under `PROTOCOL_LOCK §8` is binding; when engaged, the v2 supersession path applies the shape-neutral rubric and re-grades the sample (v2 Protocol Amendment, Cond-X).

**v1 grading completed 2026-06-22.** Final main-pass results on the N=700 stratified sample:

| Metric                          | Value                                                       |
|---------------------------------|-------------------------------------------------------------|
| Main records graded             | 700 / 700                                                   |
| Incompletes                     | 0                                                           |
| Double-grade pass1              | 70 / 70                                                     |
| Double-grade pass2              | 70 / 70                                                     |
| Total Phase 8 API spend         | \$58.82                                                     |
| Cumulative Phase 8 spend        | ~\$559                                                      |
| Main-pass min pairwise $\kappa$ | 0.296 (Sonnet-Gemini on <code>mission_intent_match</code> ) |
| PROTOCOL_LOCK §8 threshold      | 0.40                                                        |
| v2 supersession engaged         | yes (Cond-X regrade)                                        |

Main-pass pairwise  $\kappa$ :

| Outcome                     | GPT-4o $\times$ Sonnet | GPT-4o $\times$ Gemini | Sonnet $\times$ Gemini |
|-----------------------------|------------------------|------------------------|------------------------|
| <code>mission_intent</code> | 0.331                  | <b>0.709</b>           | <b>0.296</b>           |
| <code>gap_class</code>      | 0.677                  | 0.449                  | 0.713                  |

Per-outcome Krippendorff’s  $\alpha$  (pass1 double-grade) splits the ensemble into reliable and unreliable outcomes:

| Outcome                           | $\alpha$ |
|-----------------------------------|----------|
| <code>gap_classification</code>   | 0.701    |
| <code>minimum_coverage</code>     | 0.612    |
| <code>target_coverage</code>      | 0.541    |
| <code>constraint_coverage</code>  | 0.529    |
| <code>mission_intent_match</code> | 0.299    |
| <code>fabrication_count</code>    | 0.216    |
| <code>trace_completeness</code>   | 0.194    |

Objective outcomes (`gap_classification`, `minimum_coverage`, coverages) cleared or approached the threshold; subjective outcomes (`mission_intent_match`, `trace_completeness`, `fabrication_count`) fell below it.

Per-system ensemble summary (100 records per system):

| System                       | Mission | Min cov | Tgt cov | Con cov | Gap TP | Sonnet $\Delta$ |
|------------------------------|---------|---------|---------|---------|--------|-----------------|
| <code>baseline_1</code>      | 1.00    | 0.88    | 0.68    | 0.84    | 100    | +0.04           |
| <code>baseline_2</code>      | 0.26    | 0.27    | 0.17    | 0.27    | 5      | +0.06           |
| <code>baseline_3</code>      | 1.00    | 0.96    | 0.70    | 0.91    | 100    | +0.08           |
| <code>baseline_4</code>      | 1.00    | 0.89    | 0.66    | 0.84    | 100    | +0.05           |
| <code>baseline_5</code>      | 1.00    | 0.85    | 0.64    | 0.85    | 100    | +0.07           |
| <code>baseline_6</code>      | 1.00    | 0.88    | 0.64    | 0.83    | 100    | +0.04           |
| <code>mandate_primary</code> | 0.94    | 0.18    | 0.07    | 0.01    | 0      | <b>−0.30</b>    |

**Interpretation of the v1 results: rubric measures shape conformance, not semantic coverage.** The v1 per-system table reports grading against an array-of-dimensions ground-truth shape. Canonical MANDATE emits these fields as free-form objects, so the v1 cross-system row measures shape conformance rather than semantic coverage. The  $-0.30$  Sonnet delta against `mandate_primary` (versus +0.04 to +0.08 for every baseline) is a signature of shape-bias against

the canonical MANDATE schema (Finding 6, Section 4.6): the same judge that scored every other system higher than its peers scored MANDATE-primary 30 points lower than its peers, on a structurally valid output. The v2 pass below applies a shape-neutral rubric and re-grades the full corpus for the cross-system numerical comparison of record.

### v2 grading completed 2026-07-01: full-coverage cross-system results

The v2 grading pass applied a shape-neutral rubric to the full 12,000-record anonymized population under the pre-registered three-judge ensemble (Claude Opus 4.6, GPT-4o, Gemini 2.5 Pro):

| Metric         | Value                                             |
|----------------|---------------------------------------------------|
| Records graded | 12,000 / 12,000                                   |
| Incompletes    | 0                                                 |
| Judge ensemble | Opus 4.6, GPT-4o, Gemini 2.5 Pro (pre-registered) |
| Rubric         | v2 shape-neutral                                  |
| Grading window | 2026-06-24 through 2026-07-01                     |

Per-system ensemble scores across the full 12,000-record comparison (coverage outcomes, higher is better; fabrication count, lower is better):

Table 6: v2 shape-neutral rubric: full-coverage three-judge ensemble scores across nine systems. Coverage outcomes (Mission, Minimum, Target, Constraint, Trace) are higher-is-better; Fabrication count is lower-is-better. Baselines B4–B6 are pattern shells (Section 2.2), not literal framework installations. Bootstrap 95% CIs and Holm-corrected paired tests for the pre-registered Cond-B contrast family are reported in Section 6.1 (computed 2026-07-10).

| System                                             | n    | Mission | Min cov      | Tgt cov      | Con cov | Trace        | Fab   |
|----------------------------------------------------|------|---------|--------------|--------------|---------|--------------|-------|
| <b>mandate_primary</b> (v1)                        | 1500 | 0.987   | 0.243        | 0.227        | 0.290   | 1.973        | 0.577 |
| <b>cond_a</b> (canonical, structured)              | 1500 | 1.000   | <b>0.981</b> | <b>0.767</b> | 0.961   | <b>2.000</b> | 1.650 |
| <b>cond_b</b> (canonical + LLM Interp.)            | 1500 | 1.000   | 0.864        | 0.613        | 0.942   | <b>2.000</b> | 0.753 |
| <b>baseline_1</b> (Sonnet single-prompt)           | 1500 | 1.000   | 0.889        | 0.658        | 0.913   | 2.000        | 0.217 |
| <b>baseline_2</b> (GPT-4o single-prompt)           | 1200 | 0.357   | 0.332        | 0.175        | 0.335   | 1.087        | 0.019 |
| <b>baseline_3</b> (ReAct Claude)                   | 1200 | 0.989   | 0.968        | 0.713        | 0.971   | 0.384        | 1.135 |
| <b>baseline_4</b> (AutoGen Planner-Reviewer shell) | 1200 | 1.000   | 0.878        | 0.641        | 0.910   | 2.000        | 0.290 |
| <b>baseline_5</b> (CrewAI Sequential shell)        | 1200 | 0.992   | 0.824        | 0.645        | 0.910   | 1.986        | 0.213 |
| <b>baseline_6</b> (LangGraph Graph-Revision shell) | 1200 | 1.000   | 0.882        | 0.640        | 0.909   | 2.000        | 0.198 |

**Input-condition asymmetry between MANDATE conditions and baselines.** Before interpreting the numerical comparison, an important methodological asymmetry must be surfaced. Cond-A operates on *pre-extracted structured MissionInput* (Section 2.5); the deterministic planner in Cond-A does not see raw natural-language task text. Cond-B and all baselines B1–B6 operate on *raw natural-language* task text. Cond-A therefore receives the corpus in a form the baselines do not, and is not an apples-to-apples comparator against B1–B6. The apples-to-apples comparison for MANDATE under realistic input conditions is Cond-B vs B1–B6. Cond-A is reported here as an upper bound characterizing MANDATE’s deterministic planning behavior when upstream extraction is provided.

**Interpretation of the v2 results.** Four substantive characterizations follow from the full-coverage grading:



1. **Cond-B (canonical MANDATE with LLM-augmented Interpreter on raw text) is under the strongest coverage-baseline (ReAct) on minimum and target coverage, and exceeds five of six baselines on constraint coverage.** Cond-B’s `minimum_coverage` (0.864) is below `baseline_3` ReAct at 0.968; Cond-B’s `target_coverage` (0.613) is below `baseline_3`’s 0.713. Cond-B’s `constraint_coverage` (0.942) exceeds five of six baselines and is close to `baseline_3`’s 0.971. Under the input regime baselines actually operate in (raw natural-language text), MANDATE’s LLM-augmented pipeline is coverage-competitive but not coverage-dominant against a strong agentic baseline. Section 5.4 reports the per-domain breakdown.
2. **Cond-A (deterministic planning on pre-extracted structured input) exceeds every baseline on coverage under its distinct input regime.** Cond-A’s `minimum_coverage` (0.981) and `target_coverage` (0.767) are the highest values in the table. The pre-extraction is performed by a Claude Sonnet 4.6 **PreExtractor** stage whose extracted structures are the anchor fields the rubric grades, carried through deterministic planning; Cond-A’s coverage therefore characterizes the extraction-plus-planning pipeline when extraction is delegated to a frontier model, not the deterministic planner’s own generative contribution, and it is not a comparison against baseline systems, which do not receive equivalent input pre-processing.
3. **Trace completeness distinguishes structured-output systems from unstructured-output frameworks, but the judged score is not evidence of hash-chain structure.** Cond-A, Cond-B, `baseline_1`, `baseline_4`, and `baseline_6` all receive judged `trace_completeness` 2.000; `baseline_5` reaches 1.986; `baseline_2` (GPT-4o single-prompt) reaches 1.087 (incomplete traces); `baseline_3` (ReAct) reaches 0.384. Record inspection shows hash-linked 6-entry chain structure only in the MANDATE conditions (zero hash-trace markers in any baseline record on disk); the baselines’ 2.000 scores reflect judge leniency toward their output shape, consistent with the below-floor reliability of this outcome ( $\alpha = 0.19\text{--}0.28$ ). The judged advantage over `baseline_3` does not follow from Cond-B’s LLM-augmented planning but from Cond-B’s structured output format, and the categorical MANDATE-only hash-chain claim rests on disk verification, not on these scores.
4. **The v1 MANDATE-primary result under the v2 rubric (0.243 `minimum_coverage`) is a property of the fine-tuned Qwen3 specialists, not a property of the MANDATE framework.** Cond-A on the same task corpus under the same v2 rubric scores 0.981. Section 4.1 documents this two-regime pattern for Decomposition; the v2 grading extends it to the coverage-scoring level.

**Fabrication-coverage trade-off.** Systems with higher coverage report higher fabrication counts, since richer output content generates a larger denominator of factual claims for the fabrication metric. Cond-A’s 1.650 fabrications per record accompanies its 0.981 `minimum_coverage`; `baseline_2`’s 0.019 fabrications accompanies its 0.175 `target_coverage`. Fabrication comparison is most informative normalized by content volume; the raw `fabrication_count` column is reported here without normalization for reader inspection.

**Baseline observations.** `Baseline_2` (GPT-4o single-prompt) under-performs across every outcome, reflecting the limits of a single-prompt orchestration against the source-conditioned corpus. `Baseline_3` (ReAct Claude) matches or approaches MANDATE on the coverage outcomes but under-performs decisively on `trace_completeness` (0.384), reflecting the ReAct pattern’s inability to emit structured hash-chained trace records. Baselines 4, 5, 6 (AutoGen, CrewAI, LangGraph) cluster around `baseline_1` (single-prompt Sonnet) on coverage, suggesting that under source-conditioned

prompts the multi-agent orchestration patterns do not dominate a well-prompted single-model baseline.

### Per-domain v2 breakdown

The aggregate v2 table averages across in-domain (FIN, INT, SEC) records. The per-domain breakdown on `minimum_coverage` reveals that MANDATE’s cross-system standing is domain-conditional:

Table 7: Per-domain `minimum_coverage` under the v2 rubric,  $n = 400$  per (system, domain) cell (40 in-domain tasks  $\times$  10 seeds; the 30-task hold-out is out-of-domain and excluded from these cells; B2 records excluded from coverage scoring due to structural failure). Cond-A ties baseline\_3 on FIN and SEC, leads by 0.043 on INT.

| System                                    | FIN          | INT          | SEC          |
|-------------------------------------------|--------------|--------------|--------------|
| <code>mandate_primary</code> (v1)         | 0.238        | 0.175        | 0.294        |
| <code>cond_a</code> (structured input)    | <b>0.986</b> | <b>0.981</b> | <b>0.981</b> |
| <code>cond_b</code> (raw text + LLM)      | 0.877        | 0.819        | 0.872        |
| <code>baseline_1</code> (Sonnet)          | 0.894        | 0.847        | 0.907        |
| <code>baseline_2</code> (GPT-4o)          | 0.336        | 0.296        | 0.365        |
| <code>baseline_3</code> (ReAct)           | <b>0.986</b> | 0.938        | 0.980        |
| <code>baseline_4</code> (AutoGen shell)   | 0.891        | 0.845        | 0.899        |
| <code>baseline_5</code> (CrewAI shell)    | 0.846        | 0.762        | 0.863        |
| <code>baseline_6</code> (LangGraph shell) | 0.897        | 0.844        | 0.905        |

On FIN, baseline\_3 (ReAct) achieves 0.986, tying Cond-A exactly. On INT and SEC, Cond-A leads baseline\_3 by 0.043 and 0.001 respectively. The aggregate ‘Cond-A wins’ pattern is driven primarily by INT-domain performance and is a near-tie on FIN and SEC under the deterministic-planning Cond-A regime.<sup>1</sup>

Cond-B, the raw-text apples-to-apples MANDATE condition, is under baseline\_3 on every in-domain measure (0.877 vs 0.986 FIN; 0.819 vs 0.938 INT; 0.872 vs 0.980 SEC). Under raw-text input, ReAct outperforms MANDATE on coverage; MANDATE’s structural advantage lies elsewhere (trace completeness, output format, governance-signal emission).

**Reviewer-objection status.** The v2 comparative results address the substantive Reviewer R3 objections raised at the published-paper submission cycle: comparative evaluation against six pre-specified baselines is reported at full 12,000-record coverage under a shape-neutral rubric. The results characterize MANDATE’s coverage performance against baselines under two input regimes (structured Cond-A vs raw-text Cond-B) rather than claiming a single-metric win, and document that MANDATE emits structured trace-chain provenance that ReAct does not, an output-format difference between the two systems, not a coverage-quality difference. The v1 `mandate_primary` result is preserved as the fine-tune measurement and as the empirical basis for Finding 6 (LLM-judge shape bias under the v1 rubric).

<sup>1</sup>Per-domain and aggregate values are computed independently at the record level; the equal-weight mean of Cond-A’s three in-domain cells (0.9827) differs from the record-level aggregate (0.981 at  $n = 1500$ ) because the aggregate additionally includes the 300 out-of-domain hold-out records and weights by record count rather than by domain, not because of per-record variance alone.

## 6 Pre-Phase-6 Demo Evidence

The MANDATE-primary evaluation built up from three real-world demo scenarios before committing to the Phase 6 main matrix. Each demo scenario was run through three apparatus variants. The cross-variant comparison is the demo-era evidence that the Interpreter and Binding content-tripwires are driven by chunk shape rather than the user-facing request paragraph (which is identical across variants for a given scenario).

### 6.1 Pairwise Contrasts with Uncertainty (2026-07-10)

The v2 Protocol Amendment’s statistical-power section pre-registers pairwise system contrasts with bootstrap confidence intervals (10,000 resamples) and Cohen’s  $d$  on per-record outcome scores; the v1 analysis plan (Notebook 3) contributes the paired task-level Wilcoxon test and Holm–Bonferroni family-wise correction. Both are now executed by `code/scripts/bootstrap_contrasts.py` in the deposit (seed 20260710; results in `analysis/bootstrap_contrasts_results.json`).

Scope and method. The contrast family is Cond-B (the apples-to-apples MANDATE condition) against each baseline on four outcomes (minimum, target, and constraint coverage, plus trace completeness), 24 contrasts in total, restricted to the shared 120-task main corpus because baselines B2–B6 have no graded hold-out records. Two bootstrap variants are reported: the pre-registered per-record percentile bootstrap, and a task-clustered variant that resamples task IDs so seed replicates stay together (records within a task are not independent). The Wilcoxon test operates on the 120 paired task means; Holm correction is applied across all 24  $p$ -values.

Table 8: Cond-B minus baseline contrasts on the shared 120-task main corpus.  $CI_r$  = per-record percentile bootstrap 95% CI (pre-registered method);  $CI_t$  = task-clustered 95% CI;  $d$  = Cohen’s  $d$  on per-record scores;  $p_H$  = Holm-corrected two-sided Wilcoxon signed-rank on paired task means. Zero-variance rows (both systems constant at trace ceiling 2.000) have undefined  $d$ , shown as —.

| Outcome | Baseline | $\Delta$ | $CI_r$           | $CI_t$           | $d$   | $p_H$                 |
|---------|----------|----------|------------------|------------------|-------|-----------------------|
| min cov | B1       | −0.026   | [−0.036, −0.017] | [−0.044, −0.009] | −0.23 | 0.023                 |
| min cov | B2       | +0.524   | [+0.513, +0.534] | [+0.500, +0.546] | +3.92 | $< 10^{-19}$          |
| min cov | B3       | −0.112   | [−0.122, −0.102] | [−0.129, −0.095] | −0.91 | $< 10^{-16}$          |
| min cov | B4       | −0.022   | [−0.032, −0.013] | [−0.039, −0.005] | −0.19 | 0.090                 |
| min cov | B5       | +0.032   | [+0.022, +0.043] | [+0.012, +0.055] | +0.24 | 0.030                 |
| min cov | B6       | −0.026   | [−0.035, −0.017] | [−0.043, −0.009] | −0.22 | 0.023                 |
| tgt cov | B1       | −0.037   | [−0.053, −0.021] | [−0.062, −0.011] | −0.19 | 0.039                 |
| tgt cov | B2       | +0.431   | [+0.415, +0.447] | [+0.402, +0.461] | +2.15 | $< 10^{-19}$          |
| tgt cov | B3       | −0.106   | [−0.124, −0.088] | [−0.132, −0.081] | −0.48 | $< 10^{-10}$          |
| tgt cov | B4       | −0.034   | [−0.050, −0.018] | [−0.060, −0.009] | −0.17 | 0.063                 |
| tgt cov | B5       | −0.038   | [−0.054, −0.021] | [−0.064, −0.012] | −0.18 | 0.022                 |
| tgt cov | B6       | −0.034   | [−0.050, −0.018] | [−0.058, −0.009] | −0.17 | 0.063                 |
| con cov | B1       | +0.028   | [+0.021, +0.034] | [+0.016, +0.039] | +0.32 | $2.5 \times 10^{-4}$  |
| con cov | B2       | +0.605   | [+0.595, +0.614] | [+0.583, +0.627] | +5.03 | $< 10^{-19}$          |
| con cov | B3       | −0.032   | [−0.039, −0.024] | [−0.045, −0.015] | −0.34 | $3.2 \times 10^{-11}$ |
| con cov | B4       | +0.029   | [+0.023, +0.036] | [+0.018, +0.041] | +0.35 | $7.8 \times 10^{-5}$  |
| con cov | B5       | +0.030   | [+0.022, +0.038] | [+0.014, +0.051] | +0.30 | $3.8 \times 10^{-4}$  |
| con cov | B6       | +0.031   | [+0.024, +0.038] | [+0.019, +0.043] | +0.36 | $1.8 \times 10^{-5}$  |
| trace   | B1       | 0.000    | [0.000, 0.000]   | [0.000, 0.000]   | —     | 1                     |
| trace   | B2       | +0.913   | [+0.893, +0.932] | [+0.873, +0.951] | +3.68 | $< 10^{-19}$          |
| trace   | B3       | +1.616   | [+1.586, +1.645] | [+1.558, +1.671] | +4.32 | $< 10^{-19}$          |
| trace   | B4       | 0.000    | [0.000, 0.000]   | [0.000, 0.000]   | —     | 1                     |
| trace   | B5       | +0.014   | [+0.006, +0.024] | [+0.000, +0.042] | +0.13 | 1                     |
| trace   | B6       | 0.000    | [0.000, 0.000]   | [0.000, 0.000]   | —     | 1                     |

Interpretation stays within the paper’s bounded claim. Cond-B’s minimum-coverage deficit against baseline\_3 is real and large ( $d = -0.91$ ); its constraint-coverage advantage over baseline\_1 and the three pattern shells is real and small; the trace-completeness gaps against baseline\_3 and baseline\_2 are among the largest effects in the family (the single largest is the constraint-coverage advantage over baseline\_2,  $d = +5.03$ ). The descriptive Cond-A vs baseline\_3 minimum-coverage difference on the same corpus is +0.015 with task-clustered CI [+0.004, +0.029]: the interval excludes zero but lies entirely below the pre-registered minimum detectable effect ( $d = 0.25$ ), so the supplement continues to treat that boundary as effect-size-sensitive rather than substantive, and the input-condition asymmetry (Section 11) applies regardless.

## 6.2 The Three Scenarios

**Volt Typhoon (security ops).** CISO request for defense capability against the PRC state-sponsored intrusion campaign documented in CISA AA23-144A and AA24-038A. 12 source files including the CISA joint advisories (PDF), MITRE ATT&CK G1017 group page (HTML), Microsoft Threat Intelligence original disclosure blog post (HTML), DOJ KV Botnet press release (HTML), FBI search warrant affidavit 24-5018 (PDF), Wikipedia coverage, NIST SP 800-53/61. Mixed PDF + HTML.

**CrowdStrike Outage (security ops).** CIO request explicitly demanding “three distinct strategic options for endpoint security architecture” after the 19 July 2024 Falcon Sensor outage. 13 source files including the CrowdStrike Channel File 291 RCA (PDF), Microsoft recovery guidance (HTML), CISA advisory (HTML), SEC 8-K filings from Delta and United (HTML), House Homeland Security Committee testimony (HTML), NIST SP 800-218 deltas (**DOCX**), NIST CSF v1.1 presentation (**PPTX**), Intro to Framework Roadmap (PPTX). Multi-format ingest: PDF + HTML + DOCX + PPTX.

**Silicon Valley Bank Collapse (financial reporting).** CFO request explicitly demanding “three distinct strategic options for ERM and treasury overhaul” after the March 2023 Silicon Valley Bank failure. 11 source files including the Federal Reserve SVB Review April 2023 (PDF), Senate Banking Becker testimony (PDF), Federal Reserve and FDIC press releases (HTML), SEC EDGAR 10-K filings (HTML), NIST SP 800-218 / 800-53 deltas (DOCX), NIST CSF v1.1 and Roadmap presentations (PPTX), Wikipedia coverage.

### 6.3 Three Apparatus Variants Per Scenario

- **output (deterministic apparatus, no LLM):** The MANDATE pipeline runs without LLM role calls; deterministic rules produce a baseline mandate-as-code artifact for sanity. No Ollama dependency.
- **output\_ollama (Ollama mode, mcp web\_fetch chunks):** The full six-role Ollama pipeline against scenario-specific RAG indices built from text extracted by `mcp web_fetch` text-mode rendering.
- **output\_ollama\_from\_binaries (Ollama mode, binary extraction chunks):** The same six-role pipeline against scenario-specific RAG indices built from the original PDF/DOCX/PPTX binaries extracted by the apparatus’s production `pypdf`, `python-docx`, and `python-pptx` extractors.

The cross-variant comparison isolates the chunk-content variation: same scenario task text, same six fine-tuned models, same per-role temperatures, same prompts; only the chunks fed to the Procedure-role retriever differ between variants.

### 6.4 Cross-Variant Demo Results

| Scenario     | Variant         | OK | Wall (s) | LLM fb.  | $n_{COAs}$ | Fb. roles      |
|--------------|-----------------|----|----------|----------|------------|----------------|
| volt_typhoon | output          | T  | 0.0      | F        | 1          | —              |
| volt_typhoon | ollama          | T  | 238.5    | F        | 1          | —              |
| volt_typhoon | ollama_binaries | T  | 166.7    | F        | 1          | —              |
| crowdstrike  | output          | T  | 0.0      | F        | 1          | —              |
| crowdstrike  | ollama          | T  | 212.0    | F        | 1          | —              |
| crowdstrike  | ollama_binaries | T  | 172.0    | F        | 1          | —              |
| svb_collapse | output          | T  | 0.0      | F        | 1          | —              |
| svb_collapse | ollama          | T  | 195.8    | F        | 1          | —              |
| svb_collapse | ollama_binaries | T  | 215.1    | <b>T</b> | 1          | <b>Binding</b> |

All nine demo runs produce  $n_{COAs} = 1$ , reaffirming the Decomposition single-COA prior at the demo level. The only LLM fallback in the demo work is the SVB from-binaries Binding refusal (Section 6.8 below).

## 6.5 The Interpreter Mode Flip Cross-Variant

The verbatim `anchor.minimum.description` per scenario per variant. This is the demo-era demonstrative result for the content-tripwire mode flip: the request paragraph is identical across variants; the chunks fed to the upstream Procedure retriever differ; the Interpreter output mode flips accordingly.

**Volt Typhoon (stable: det-prefix in all three variants).**

```
output:
  Minimally satisfy: Team, this is the CISO speaking. CISA
  has now published two binding-direction advisories
  (AA23-144A and AA24-038A) ...

output_ollama:
  Minimally satisfy: Team, this is the CISO speaking. CISA
  has now published two binding-direction advisories
  (AA23-144A and AA24-038A) ...

output_ollama_from_binaries:
  Minimally satisfy: Team, this is the CISO speaking. CISA
  has now published two binding-direction advisories
  (AA23-144A and AA24-038A) ...
```

**CrowdStrike (flips: `web_fetch` → clean, others → det-prefix).**

```
output:
  Minimally satisfy: Team, this is the CIO. As we discussed
  at the executive briefing, the board has formally
  requested a strategy memo follo...

output_ollama (web_fetch chunks):
  Three distinct strategic options for endpoint security
  architecture with (a) operational and capex
  implications, (b) residual risk profile after one year,
  (c) vendor-relationship and contractual changes, and
  (d) implementation timeline.

output_ollama_from_binaries (binary extraction chunks):
  Minimally satisfy: Team, this is the CIO. As we
  discussed at the executive briefing, the board has
  formally requested a strategy memo follo...
```

**SVB (flips in opposite direction: from-binaries → clean, others → det-prefix).**

```
output:
  Minimally satisfy: Team, this is the CFO. The audit
  committee has asked for a strategy memo following the
  Silicon Valley Bank collapse of M...

output_ollama (web_fetch chunks):
  Minimally satisfy: Team, this is the CFO. The audit
  committee has asked for a strategy memo following the
  Silicon Valley Bank collapse of M...

output_ollama_from_binaries (binary extraction chunks):
  Three distinct strategic options with operational/capex
  implications, residual risk profiles modeled against
```

SVB-class deposit outflow, contractual/personnel  
changes, and implementation timelines aligned with next  
SOX 404 cycle.

CrowdStrike and SVB *flip in opposite directions* between the web\_fetch and from-binaries variants. Same scenario; same request paragraph; same fine-tuned Interpreter model. The clean-distillation output appears on CrowdStrike under web\_fetch chunks and on SVB under from-binaries chunks. The deterministic-prefix output appears on CrowdStrike under from-binaries chunks and on SVB under web\_fetch chunks. This is the demo-era demonstrative result for the finding documented at Phase 6 scale (Section 4.2): the Interpreter mode is selected by chunk content, not by request paragraph.

## 6.6 The Binding Refusal Demo Provenance

The SVB from-binaries run triggered the Binding refusal (last row of cross-profile pilot table). The dedicated Binding-refusal diagnostic captured three consecutive direct-Ollama calls against this failing pipeline state; verbatim quotes are in Section 4.4 of this document. The diagnostic established the structured-refusal pattern that Phase 6 measured at scale (244 of 1500 records).

## 6.7 Demo Findings Validated at Phase 6 Scale

The five demo-era findings and their Phase 6 replication status:

| Demo finding                   | Phase 6 replication      | Phase 6 sample    |
|--------------------------------|--------------------------|-------------------|
| Decomposition single-COA prior | Yes (1480/1480)          | MP main + holdout |
| Interpreter mode flip          | Yes (per-domain rates)   | MP main (1180 ok) |
| Validator gap-instability      | Yes (14-19% range)       | MP main (1180 ok) |
| Binding structured refusal     | Yes (244/1500, FIN low)  | MP main + holdout |
| Intake content-tripwire        | NEW at Phase 6 (20/1200) | Two SEC tasks     |

Four of five demo findings replicated cleanly at scale. The fifth (Intake content-tripwire) emerged only at Phase 6 scale because the affected phrase “Here’s the constraint.” appears in only 2 of 120 main-corpus tasks, both outside the 6-task pilot smoke (Section 6.8).

## 6.8 Cross-Profile Pilot Findings (Pre-Phase-6, April 2026)

A six-week-prior pilot evaluation tested three apparatus configurations against the small `authorized_lab` corpus (6 cases, pentest domain) using the `ab_evaluation.py` script: *deterministic* (rule-based, no LLM), *base* (Qwen3-32b + Qwen3-8b untuned via Ollama), and *tuned* (fine-tuned *mandate-\** roles on Qwen3-32b). The aggregate per-profile results across 15 logged runs:

| Profile              | ok rate | check pass % | avg duration (s) | avg COAs | avg gaps |
|----------------------|---------|--------------|------------------|----------|----------|
| <b>deterministic</b> | 5/6     | 100.0        | 0.0              | 1.50     | 1.5      |
| <b>base</b>          | 5/6     | 90.5         | 79.7             | 1.17     | 3.2      |
| <b>tuned</b>         | 5/6     | 92.9         | 101.2            | 1.17     | 3.7      |

**Three pilot observations.** (a) Pipeline ok rate is identical at 5/6 across all three profiles: the deterministic apparatus completes 5 of 6 cases and LLM augmentation neither increases nor degrades that. One specific case fails structurally under every configuration. (b) LLM augmentation initially *degrades* check pass rate (deterministic 100%  $\rightarrow$  base 90.5%  $\rightarrow$  tuned 92.9%) before iteration-of-apparatus-fixes recovers it (tuned post-fix runs reach 100%, with one regression observed when an interpreter compaction broke checks before being rolled back). (c) LLM modes produce *fewer COAs* (1.50  $\rightarrow$  1.17) and *more gap reports* (1.5  $\rightarrow$  3.2–3.7) than the deterministic path. The under-COA observation prefigures the Phase 6 *Decomposition single-COA prior* finding (Section 4.1).

**Scope.** The pilot tested a single LLM vendor family (Qwen3 via Ollama). It does not establish cross-vendor variance; v2’s planned multi-vendor Cond-B will address that question at the 1500-task scale. Full per-run aggregates and iteration history are preserved at `standalone data results/v1_pilot_cross_profile/`; the April 2026 source logs are at `~/Desktop/AEGIS/logs/`.

## 7 Perturbation Suite

**Headline O5 result:** across the three MANDATE conditions tested against the 350-perturbation suite (Phase A, completed 2026-07-01), *prompt-injection perturbations produce zero structural failures*. MANDATE-primary emits 500/500 structurally complete records on the prompt-injection subset (Wilson 95% CI [99.24, 100.00]); Cond-A and Cond-B each emit 50/50 records (Wilson 95% CI [92.87, 100.00]) with the same result. The apparatus does not lose trace-chain completeness and does not skip COA emission under any of the three tested MANDATE conditions on this perturbation type. Structural completion of the pipeline under adversarial input is what this Phase A metric measures; semantic robustness to the injected instruction (i.e., whether the system produces output that follows the injected directive rather than the original task) is not decidable from the structural signal; its Phase B semantic grading under the v2 rubric was paused at 80.7% coverage under Deviation D-13 (see §11, §11.5). Full per-perturbation-type results are in Table 9.

The 350-perturbation suite was generated from the frozen main corpus and frozen at the `perturbation_freeze_v1` tag (commit 068fe94). The suite carries seven canonical perturbation types at 50 perturbations each. Each perturbation references a base task ID and applies a specific perturbation function to the task text.

| Perturbation type                      | Count      |
|----------------------------------------|------------|
| <code>ambiguity_injection</code>       | 50         |
| <code>contradictory_constraints</code> | 50         |
| <code>length_perturbation</code>       | 50         |
| <code>missing_required_field</code>    | 50         |
| <code>out_of_distribution_input</code> | 50         |
| <code>prompt_injection</code>          | 50         |
| <code>surface_noise</code>             | 50         |
| <b>Total</b>                           | <b>350</b> |

**Record schema.** Each perturbation record carries `perturbation_id`, `base_task_id` (the source main-corpus task), `perturbation_type` (one of seven), `sub_type` (category within a type; e.g., for `prompt_injection`: `direct_override`, `context_break`, `role_swap`), `domain`, `request_text` (the



perturbed task text the system under test sees), `generation_ms`, `input_tokens`, `output_tokens`, `model`, and an `internal_note` describing the perturbation applied.

**O5 adversarial-resistance runs (completed 2026-07-01).** Under v2 Deviation 11 (Section 9), the O5 outcome was reactivated and system runs against the 350-perturbation suite completed. MANDATE-primary generated 3,500 records (350 perturbations  $\times$  10 seeds); Cond-A and Cond-B each generated 350 records (deterministic; 1 seed per perturbation). Structural-invariance results by perturbation type:

Table 9: O5 structural-invariance results by perturbation type, MANDATE-primary + Cond-A + Cond-B. “Trace  $\geq 6$ ” counts records with a hash-chained trace of at least 6 entries (Property 2). Wilson 95% CIs bracket the ok-rate estimate; because the 10 seed-replicates per perturbation are not independent trials, the conservative basis for the MANDATE-primary rows is the 50 unique perturbation instances per type ( $n = 50$ : 100% rows give [92.87, 100.00]); the per-record  $n = 500$  intervals are anti-conservative and shown for reference only. Under a Holm-Bonferroni correction across the seven perturbation types at  $\alpha = 0.05/7$ , the substantive per-system conclusions (every ok-rate exceeds 94%) are preserved. Semantic scoring of these records under the v2 rubric (Phase B) is paused at 80.7% coverage under Deviation D-13.

| System                            | Perturbation type                      | ok-rate                 | Wilson 95% CI   | Trace $\geq 6$ |
|-----------------------------------|----------------------------------------|-------------------------|-----------------|----------------|
| <code>mandate_primary</code>      | <code>ambiguity_injection</code>       | 500/500 (100.0%)        | [99.24, 100.00] | 500/500        |
| <code>mandate_primary</code>      | <code>contradictory_constraints</code> | 470/500 (94.0%)         | [91.56, 95.77]  | 470/500        |
| <code>mandate_primary</code>      | <code>length_perturbation</code>       | 500/500 (100.0%)        | [99.24, 100.00] | 500/500        |
| <code>mandate_primary</code>      | <code>missing_required_field</code>    | 500/500 (100.0%)        | [99.24, 100.00] | 500/500        |
| <code>mandate_primary</code>      | <code>out_of_distribution_input</code> | 500/500 (100.0%)        | [99.24, 100.00] | 500/500        |
| <code>mandate_primary</code>      | <b><code>prompt_injection</code></b>   | <b>500/500 (100.0%)</b> | [99.24, 100.00] | <b>500/500</b> |
| <code>mandate_primary</code>      | <code>surface_noise</code>             | 500/500 (100.0%)        | [99.24, 100.00] | 500/500        |
| <code>cond_a</code> (all 7 types) | —                                      | 350/350 (100.0%)        | [98.91, 100.00] | 350/350        |
| <code>cond_b</code> (all 7 types) | —                                      | 350/350 (100.0%)        | [98.91, 100.00] | 350/350        |

**Reading the table.** MANDATE-primary shows one below-100% row: `contradictory_constraints` at 94% (30 of 500 records failed the pipeline on logical-contradiction inputs). The Cond-A and Cond-B canonical MANDATE paths handle contradictory constraints at 100% pipeline completion; on those conditions, contradictory inputs are handled via the Binding structured-refusal mechanism (Section 4.4) rather than by pipeline completion in the conventional sense. Semantic quality of the refusal responses under adversarial input is a Phase B evaluation outcome (paused under Deviation D-13), not a Phase A structural check.

**Phase B status at deposit closeout (2026-07-08).** Baseline perturbation generation is scoped to B1–B4 (Deviation D-12; the CrewAI and LangGraph shells are represented by the B4 multi-agent-shell class representative). B1–B3 runs are complete at 3,500 records each; B4 generation was halted at 3,021/3,500 records (86.3%) at deposit closeout under the same budget decision (D-13), leaving a generated perturbation-record pool of 17,721 of the 18,200-record scoped target. Semantic v2 grading of the 18,200-record scoped set is paused at 14,685 main-pass grades (80.7%) under Deviation D-13: complete for MANDATE-primary (3,500/3,500), Cond-A (350/350), Cond-B (350/350), and `baseline_2` (3,500/3,500); near-complete for `baseline_1` (3,498/3,500) and `baseline_3` (3,487/3,500); not started for `baseline_4`. Double-grade IRR pass 1 stands at 816 of 3,640 records; pass 2 is not started. No cross-system semantic adversarial-resistance comparison is

claimed from the partial grades; the grading layer is resumable from frozen records via `grade-v2-skip-existing` without record regeneration.

**Why seven types.** The seven types span complementary adversarial dimensions: `ambiguity_injection` adds genuinely ambiguous references; `contradictory_constraints` introduces logical contradictions between requirements; `length_perturbation` tests robustness to abnormal length (very short or very long requests); `missing_required_field` strips a critical specification element; `out_of_distribution_input` substitutes domain-irrelevant text; `prompt_injection` adds adversarial instructions inside the request; `surface_noise` adds whitespace, typos, formatting glitches. Together they probe different specification failure modes a real-world system might encounter.

## 8 Corpus Candidate Pool Audit

The deposit’s selected tasks (120 main + 30 hold-out + 6 pilot) are a subset of larger candidate pools produced by the source-conditioned generation pipeline. The unused residue is a v2 evaluation resource.

### 8.1 Main Corpus

| Main corpus state          | Count |
|----------------------------|-------|
| Total candidates generated | 262   |
| Selected for Phase 6       | 120   |
| Residue (available for v2) | 142   |

**Residue by domain:** `financial_reporting` 49, `intelligence_collection_tasking` 46, `security_operations_reporting` 47.

**Residue by category:** `full_specification` 46, `gap_triggering` 47, `stretch_case` 49.

The residue’s distribution by domain and category is roughly balanced. The DEVIATIONS.md (2026-06-10 Intake content-tripwire entry) cites this 142-candidate residue as the v2-candidate-replacement source for `TASK-MAIN-SEC-038` and `TASK-MAIN-SEC-040` (the Intake-tripwire tasks). A future evaluation could swap those tasks with alternative SEC `stretch_case` candidates from the 49-strong SEC residue pool to test whether the substantive findings hold against a different task set.

### 8.2 Hold-Out Corpus

| Hold-out corpus state      | Count |
|----------------------------|-------|
| Total candidates generated | 44    |
| Selected for Phase 6       | 30    |
| Residue (available for v2) | 14    |

The 14-candidate residue covers all three categories (`full_specification` 4, `gap_triggering` 5, `stretch_case` 5) and can be used for v2 hold-out expansion if a larger generalization sample is needed.

### 8.3 Pilot Corpus

| Pilot corpus state           | Count |
|------------------------------|-------|
| Total candidates generated   | 183   |
| Selected for pilot scaffolds | 6     |
| Residue (available for v2)   | 177   |

The pilot pool is much larger than the selected pilot subset (183 vs 6). A future v2 evaluation that wants a larger pilot sample (for example, 30 tasks across the three main-corpus domains for a faster smoke test before the full 120-task main run) has 177 unused candidates available.

## 9 Deviation Table

Following Willroth & Atherton (2024) recommended structured format, all v1- and v2-cycle deviations from pre-registration are recorded below as a single table to support reviewer cross-comparison. Each deviation entry records: date, type, the pre-registered plan, the actual change with rationale, severity, and the substantive effect on conclusions. The full chronological narrative for each entry is preserved at `code/00_preregistration/DEVIATIONS.md` in the deposit. Forward-looking v2 protocol amendments (introduction of three measured conditions, restored full-coverage Opus grading, reactivated O5 perturbation outcome) are documented separately in the companion **v2 Protocol Amendment** document bundled with the v2 deposit.

| # | Date       | Type                      | Pre-registered plan                                           | Actual / rationale                                                                                                                                                                                                                                                                                                                                       | Severity | Effect on conclusions                                                                                                                                                                                                                                               |
|---|------------|---------------------------|---------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1 | 2026-06-04 | Scope: ground-truth cycle | SME realism audit + anchor accept/edit/reject within v1 cycle | SME ratification cycle scheduled with v2 full-coverage grading but <i>not executed</i> before the D-13 budget pause (open item); v1 ground truth comprises Claude Opus 4.6 scaffolds against curated public-source chunks; <code>gt_freeze_v2</code> layers cleanly atop v1 baseline                                                                     | Scope    | v1 $\kappa$ measures inter-judge agreement on the v1 ground-truth answer key (§2.4); all reported v2 grades use the unratified model-authored scaffolds; SME validation remains open, resumable per D-13                                                            |
| 2 | 2026-06-04 | Scope: ablation phasing   | Seven pre-registered ablations (A1–A7)                        | A3 ( <code>no_gap_analysis</code> ) and A5 ( <code>no_registry</code> ) were pre-registered as apparatus-configuration ablations. A3 ran cleanly at 1500-record scale; A5 was attempted at the same scale and empirically characterizes an apparatus-level invariant property (Finding 7), reclassifying A5 to source-level alongside A1, A2, A4, A6, A7 | Scope    | v2 ablation analysis reports A3 at 1500-record scale with the pre-registered structural check satisfied. A5 reports the apparatus-level invariant property (Section 4.7); the source-level form of A5 ships with the upstream variant tags alongside A1/A2/A4/A6/A7 |

| # | Date                     | Type                    | Pre-registered plan                                                                                                                   | Actual / rationale                                                                                                                                                                                        | Severity    | Effect on conclusions                                                                                                                                                                                                                     |
|---|--------------------------|-------------------------|---------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 3 | 2026-06-10               | Apparatus               | Watchdog rule: halt on <code>wall_clock_ms &lt; 60000</code>                                                                          | Refined: halt on <code>wall_clock_ms &lt; 60000</code> AND all roles report <code>llm_used=False</code>                                                                                                   | Minor       | Phase 6 O4 measurement integrity improved; legitimate Intake failures no longer over-quarantined                                                                                                                                          |
| 4 | 2026-06-13               | Apparatus               | Hold-out runs produce clean RunRecords                                                                                                | 300 contaminated records (Ollama outage; detected via <code>all-roles llm_fallback=True</code> signature) quarantined and regenerated cleanly                                                             | Minor       | 0/300 contamination in regenerated hold-out; <code>outputs_freeze_v1</code> preserved, <code>outputs_freeze_v1_1</code> cut at the corrected commit                                                                                       |
| 5 | 2026-06-16               | Finding promotion       | Phase 6 measurement only                                                                                                              | Intake content-tripwire on 20/1200 records characterized as substantive finding (Section 4.5)                                                                                                             | Substantial | Reproducible across all 10 seeds per task; trigger is the phrase “Here’s the constraint:”, not the source document. Six of eight NIST SP 800-115-derived tasks show 0/10 Intake failure rate; only the 2 carrying the trigger phrase fail |
| 6 | 2026-06-04 (pre-deposit) | Scope: O5 phasing       | O5 adversarial outcome measured against the 350-perturbation suite                                                                    | Suite generated and frozen at <code>perturbation_freeze_v1</code> ; O5 runs scheduled with v2 full-coverage grading cycle                                                                                 | Scope       | O5 outcome reports with v2 (see v2 Protocol Amendment)                                                                                                                                                                                    |
| 7 | 2026-06-17               | Apparatus robustness    | Three-judge ensemble grading produces per-record checkpoints                                                                          | Grader patched to per-record checkpointing with <code>-skip-existing</code> resume + bounded judge concurrency; four regression tests guard the patched behavior                                          | Minor       | No methodology change; apparatus robustness improved                                                                                                                                                                                      |
| 8 | 2026-06-18               | Phase 8 sampling design | Full-coverage three-judge grading across all 9000 anonymized RunRecords; Claude Opus 4.6 as the Anthropic judge; 20% double-grade IRR | v1 cycle ran a pre-registered stratified random sample $N=700$ ( $100$ per system $\times 7$ ) with Sonnet substituted for Opus and 10% IRR; v2 cycle restores the full-coverage Opus design with 20% IRR | Refinement  | Five substantive findings (§4) evidenced at full coverage. v2 supersedes for the cross-system numerical comparison under the full-coverage design (see v2 Protocol Amendment)                                                             |

| #  | Date       | Type                    | Pre-registered plan                                                                                                           | Actual / rationale                                                                                                                                                                                                                                                                                                                                                                                                   | Severity    | Effect on conclusions                                                                                                                                                                                                                                                                                                                                                                                                                     |
|----|------------|-------------------------|-------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 9  | 2026-06-23 | Methodology refinement  | v1 Phase 8 grading rubric expected to <code>anchor.minimum/target</code> as arrays of {dimension, threshold, rationale} dicts | v2 introduces a shape-neutral rubric matched to canonical MANDATE's free-form-object schema (verified against canonical <code>mlt-stack 1.0.0rc1</code> : same shape as v1 records on same task inputs). v2 measures three conditions (Cond-X regrade, Cond-A pre-extracted structured input, Cond-B LLM-augmented end-to-end) under the shape-neutral rubric                                                        | Refinement  | v2 is the source of cross-system claims for citation. v1 substantive content-tripwire findings (§4) derive from on-disk RunRecord inspection and hold under both v1 and v2 rubrics                                                                                                                                                                                                                                                        |
| 10 | 2026-06-23 | Methodology restoration | D-08 substitution of Sonnet for Opus as the Anthropic judge and reduction of double-grade IRR from 20% to 10%                 | v2 grading restores the pre-registered ensemble: Claude Opus 4.6 as the Anthropic judge, GPT-4o unchanged, Gemini 2.5 Pro unchanged; the restored design also specifies 20% double-grade IRR at full 12,000-record coverage                                                                                                                                                                                          | Restoration | v2 main-pass grading (completed 2026-07-01, 12,000/12,000) reports the cross-system comparison under the pre-registered judge configuration and supersedes v1 for cross-system citation. The v2 20% double-grade IRR component was not executed before the D-13 budget pause; v2 reliability is evidenced by the v1 IRR artifact (Section 5.4) and the shape-neutral rubric change, and the v2 double-grade remains a resumable open item |
| 11 | 2026-06-23 | Methodology restoration | D-06 deferral of O5 adversarial-resistance outcome measurement against the 350-perturbation suite                             | O5 reactivated: the 350-perturbation suite runs against the three MANDATE conditions (Phase A, complete 2026-07-01: 3,500 + 350 + 350 records) and against baselines under the Phase B scope (D-12). MANDATE-condition structural invariance verified: 100% ok-rate on six of seven perturbation types for MANDATE-primary (94% on <code>contradictory_constraints</code> ); 100% on all seven for Cond-A and Cond-B | Restoration | O5 structural-invariance outcome reported in Section 7; semantic grading of perturbation records under v2 rubric initiated as Phase B (paused under D-13)                                                                                                                                                                                                                                                                                 |

| #  | Date       | Type                          | Pre-registered plan                                                                                               | Actual / rationale                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Severity       | Effect on conclusions                                                                                                                                                                                                                                                                                                                                                                                                                    |
|----|------------|-------------------------------|-------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 12 | 2026-07-06 | Scope: Phase B baseline set   | O5 perturbation runs across all six baselines (B1-B6)                                                             | Perturbation generation scoped to B1-B4 (HANDOFF_24c). B4-B6 are implementation-pattern shells sharing the same LLM (Claude Sonnet) and architectural class (PROTOCOL_LOCK §2.2; empirically confirmed by the 2026-07-01 literature-comparability audit); baseline_4 (AutoGen shell) retained as the multi-agent-shell class representative. Estimated saving ~\$4,200 API spend / ~5.8 days wall-clock                                                                                                                                                                                                                                                                        | Scope          | Baseline archetype coverage preserved: single-prompt Anthropic (B1), single-prompt OpenAI (B2), agentic reasoning (B3 ReAct), multi-agent orchestration shell (B4 as class representative). CrewAI-shell and LangGraph-shell perturbation behavior is not separately measured under O5                                                                                                                                                   |
| 13 | 2026-07-08 | Scope budget: Phase B grading | + Full-coverage semantic v2 grading of the scoped 18,200-record perturbation set (D-12) with 20% double-grade IRR | Grading paused at 14,685/18,200 main-pass grades (80.7%); double-grade IRR pass 1 at 816/3,640, pass 2 not started. Baseline_4 perturbation generation halted at 3,021/3,500 records (86.3%) at the same closeout (B1-B3 complete at 3,500 each; generated pool 17,721 of the 18,200 scoped target). Cumulative multi-provider API spend exceeded the pre-registered projection: the per-grade cost basis in the projection was Sonnet-priced and was not re-derived when D-10 restored Opus as the Anthropic judge at ~5× per-call cost. Frozen records and completed grades preserved; grading resumable via <code>grade-v2-skip-existing</code> without record regeneration | Scope + budget | No Phase B semantic claims are made in this deposit. Phase A structural-invariance results (Section 7), including 100% prompt-injection structural completeness across all three MANDATE conditions, are complete and independent of the grading layer. MANDATE-condition semantic perturbation grades are complete (4,200/4,200); the paused remainder is baseline-side and IRR grading. Ungraded records preserved for future re-grade |

**Forward-compatibility statement.** All thirteen deviations are forward-compatible with v2 re-evaluation and re-grade. The Phase 6 RunRecords at `outputs_freeze_v1_1`, the corpus at

`corpus_freeze_v1`, the 350-perturbation suite at `perturbation_freeze_v1`, and the anchor scaffolds at `gt_freeze_v1` can all be re-graded against an SME-revised ground truth, replayed under canonical MANDATE adapters, or extended with the deferred ablations without regenerating the v1 measurement infrastructure. The CC BY 4.0 license on this v1 record permits the v2 deposit to layer cleanly on top.

## 10 From-Binaries Demo: Apparatus Validation via Source-Binary Replay

This section consolidates the substantive findings from the three demo-era source-binary replays that grounded the Interpreter mode flip (Finding 2) and the Binding structured refusal (Finding 4). The pre-Phase-6 demos used three real-world scenarios (Volt Typhoon CISO request, CrowdStrike CIO outage response, SVB CFO crisis briefing) re-ingested from the original PDF / DOCX / PPTX / HTML binaries through the apparatus’s production extractors.

**Cross-extraction-path evidence.** For each scenario the demo work compared three apparatus variants: web-fetch text-mode extracts (original), from-binaries production-extractor pass, and a direct-Ollama role-prompt replay. Volt Typhoon and CrowdStrike completed cleanly across variants. SVB exhibited the Binding refusal phenomenon only on the from-binaries variant: same scenario, same fine-tunes, only the upstream chunk shape differed between the clean run and the refusal-emitting run. This established that the content-tripwire is driven by chunk shape upstream of the role, not by the user-facing request paragraph (which is byte-identical across variants). The Interpreter mode flip evidence has the same cross-variant signature.

**Direct-role-prompt captures of the Binding refusal.** The apparatus discards raw model responses after three parse retries; a targeted diagnostic reconstructed the exact prompt the apparatus sent to `mandate-binding` on the failing SVB run and called Ollama directly three times against the captured prompt. All three captured payloads were clean valid JSON with a single field `error` containing a natural-language explanation of why the role declined to bind. The verbatim payloads are reproduced in Section 4.4. These captures provide the direct evidence that the Binding fine-tune declines to bind contradictory inputs, not failing at JSON emission.

**Demo hypotheses vs Phase 6 evidence.** At demo time the SVB Binding refusal phenomenon was understood as “something about financial-domain chunks specifically.” The Phase 6 main-matrix evidence inverts that hypothesis: financial is the lowest refusal-rate domain at 2.0%, and intelligence-collection is the highest at 40.2%. The demo hypothesis was domain-specific; the Phase 6 evidence is domain-comparative and contradicts the demo-era reading. This is documented in Section 4.4 and motivates the candidate refusal-as-gap patch (Section 9).

## 11 Threats to Validity

Following the standard construct-internal-external-conclusion validity taxonomy (Cook & Campbell), extended here with a fifth subsection covering instrumentation and temporal validity given the two-rubric evolution across this evaluation, the substantive threats to the empirical claims in this supplement are enumerated below with the corresponding mitigation or acknowledgment.

## 11.1 Construct validity

- **“Structural validity” is a bounded construct.** Section 1.2 Claim 1 defines structural validity explicitly as apparatus-produced schema-conforming mandate-as-code with resolvable trace hashes. The construct does not measure content correctness; correctness is addressed at the coverage-scoring level in Section 5.4 and in the per-Finding evidence in Section 4.
- **Judge-based grading operationalizes coverage.** The v2 per-outcome ensemble scores (Section 5.4) treat three-judge ensemble means as coverage estimates. Per-outcome Krippendorff’s  $\alpha$  under the v1 rubric shows subjective outcomes (`mission_intent_match`  $\alpha = 0.299$ , `trace_completeness`  $\alpha = 0.194$ , `fabrication_count`  $\alpha = 0.216$ ) below the conventional 0.667 floor for tentative interpretation. Objective outcomes meet or approach the floor depending on the basis: on the pass-1 double-grade ( $n = 70$ ), `gap_classification`  $\alpha = 0.701$  and `minimum_coverage`  $\alpha = 0.612$ ; on the main pass ( $n = 700$ ), `gap_classification`  $\alpha = 0.597$  and `minimum_coverage`  $\alpha = 0.697$ ; neither basis clears the floor on both outcomes simultaneously. Subjective-outcome values in the v2 table should be interpreted with the caveat that ensemble means average judgments that disagree; the v2 rubric refinement targeted these outcomes specifically.
- **Ground-truth authoring by Claude Opus 4.6 creates a measure-criterion entanglement.** v1 ground-truth scaffolds were authored by Claude Opus 4.6; Claude Opus 4.6 also serves as one of the three v2 judges. This introduces a measure-criterion loop that scoring outcomes against the authoring model itself does not break. The SME ratification cycle scheduled with v2 grading (Section 9 Deviation 1) is the substantive resolution path; it was not executed before the D-13 budget pause and remains an open item, so all grades reported here carry this entanglement unresolved.

## 11.2 Internal validity

- **Judge-baseline overlap risk.** Baseline B1 uses Claude Sonnet 4.6; the Anthropic judge in v1 was Claude Sonnet 4.6. Under v2 (Deviation 10) the Anthropic judge is Claude Opus 4.6, mitigating same-model self-grading bias for B1’s scoring pass. Baselines B4–B6 also use Claude Sonnet; the v2 Opus judge is compositionally distinct. The overlap risk is two-legged, however: Claude Opus 4.6 also authored the ground-truth scaffolds (see the construct-validity item above), so the v2 judge swap removes the same-model leg for the Sonnet baselines while leaving the judge-as-criterion-author leg in place for Opus.
- **Anonymization is field-set-observable.** Records are anonymized by identifier substitution before grading. However, MANDATE outputs are trivially identifiable by their field set (`trace`, `gap_reports`, `registry_reference`) and by the invariant COA name “Minimal manual assessment approach” in the v1 fine-tuned outputs. Judges may thus recognize MANDATE records structurally even when identifier-anonymized. This is a known limitation of behavioral anonymization for structurally distinctive systems.
- **Cond-A vs baselines input asymmetry.** Section 5.4 explicitly discloses that Cond-A receives pre-extracted structured `MissionInput` while baselines B1–B6 receive raw natural-language text. Cond-B is the apples-to-apples MANDATE comparator against baselines under the raw-text input regime. This asymmetry is surfaced in the Section 5.4 interpretation paragraph.
- **Cross-vendor LLM roles are not uniformly firing.** Section 1.2 Claim 3 discloses that on Llama 3.2 and Phi-3, the any-role LLM fallback union reaches 100% of records (Interpreter-only



schema-validation failures are 64% and 20%, respectively; the remainder of the union comes from the other roles); the deterministic fallback path produces the structural completeness on those vendors. Cross-vendor structural invariance therefore demonstrates graceful degradation of the apparatus under per-role LLM failure across LLM capability tiers, not LLM-role invariance across all four vendors uniformly.

- **Pretraining-contamination check is one-sided.** Corpus- contamination checks in this evaluation cover hold-out authoring (no in-corpus task text was seen by the systems under test before Phase 6). They do not cover *pretraining* exposure: Claude Opus 4.6 authored the ground-truth scaffolds and serves as one of the three v2 judges, so overlap between the 18 public source documents (NIST SP 800-115, COSO ERM, PCAOB AS 2201, MITRE ATT&CK, ODNI ICD 203, etc.) and Opus 4.6’s pretraining corpus cannot be excluded. This is a boundary condition of any evaluation whose corpus and judge are both instantiated by frontier LLMs.

### 11.3 External validity

- **Four domains, all governance-family.** The 120-task main corpus spans three in-domain corpora (`security_operations_reporting`, `financial_reporting`, `intelligence_collection_tasking`) plus one out-of-domain hold-out (`software_engineering_specification`). All four are structured governance-family tasks. Generalization to substantially different task families (e.g., open-domain generation, creative writing, general-purpose Q&A) is not supported by this evaluation.
- **Fine-tunes are Qwen3-only.** The v1 MANDATE-primary role specialists are fine-tuned on Qwen3-8B (Intake, Procedure) and Qwen3-32B (Interpreter, Decomposition, Binding, Validation). Cross-LLM-family evidence applies only to canonical MANDATE (Cond-A, Cond-B), not to the fine-tuned specialists. Any substantive claim for v1 `mandate_primary` numbers is therefore properly restricted to the Qwen3-fine-tuned regime.
- **Sample-size and corpus-provenance boundaries.** The 120-task main corpus is Opus-authored at temperature 0 from 18 curated public source documents. Task selection from the 262-task candidate pool (Section 8) removed 142 candidates via the process documented in the corpus residue inventory. Statistical inference from this sample to the population of “all governance-domain mission requests” is not made.
- **No human-expert coverage baseline.** The v2 grading table (Section 5.4) compares nine automated systems against Opus-authored ground-truth scaffolds; it does not carry a human-expert upper bound. The 30-specification human expert upper bound is disclosed in Section 9 as an external-dependency-bound deviation subject to practitioner availability. Coverage claims should be read as system-to-system comparisons rather than as absolute coverage relative to expert practice.

### 11.4 Instrumentation and temporal validity

- **Rubric evolution v1  $\rightarrow$  v2 is an instrumentation threat.** The v1 grading rubric expected an array-of-dimensions `anchor.minimum/target` shape; the v2 rubric is shape-neutral and matched to canonical MANDATE’s free-form-object schema (Section 4.6). The rubric change during evaluation is an instrumentation-of-measure threat: measurements taken with the two instruments are not directly comparable, and cross-system rankings can shift with the instrument (as they do between the v1 per-system table in §5.4 and the v2 per-system table). The threat is mitigated

by publishing both v1 and v2 tables and by preserving v1 as the empirical basis for Finding 6 (LLM-judge shape bias) rather than as a cross-system comparison.

- **Baseline authenticity.** Baselines B4, B5, and B6 are pattern shells (Section 2.2) implementing the orchestration shapes of AutoGen, CrewAI, and LangGraph in apparatus code rather than the literal library installations. Comparison to the literal frameworks is out of scope for this evaluation and should not be inferred from B4–B6 columns. Baselines B1 (Claude Sonnet single-prompt), B2 (GPT-4o single-prompt), and B3 (ReAct Claude) are direct implementations without this abstraction.
- **Temporal validity.** All 2026Q2 data were collected between 2026-06-04 (corpus freeze) and 2026-07-01 (Stage 4 grading completion). Frontier LLM APIs (Claude Sonnet 4.6, Claude Opus 4.6, GPT-4o, Gemini 2.5 Pro) may snapshot-drift between this evaluation period and later replication attempts; API endpoints and model versions are pinned in the deposit’s `judges_config.json` but the underlying model weights are provider-controlled. Long-term replication reliability depends on provider version stability.

## 11.5 Conclusion validity

- **Confidence intervals on v2 ensemble means: delivered for the contrast family (method pre-registered; family enumerated at analysis time, 2026-07-10).** The v2 grading table (Section 5.4) reports point estimates; Section 6.1 adds per-record and task-clustered bootstrap 95% CIs plus Holm-corrected paired Wilcoxon tests for the 24 Cond-B contrasts (bootstrap method and MDE pre-registered in the v2 amendment; the specific contrast enumeration, tests, and seed were fixed at analysis time, after unblinding, and inference is labeled accordingly). The strongest apples-to-apples comparison, Cond-B vs `baseline_3` on `minimum_coverage`, is  $\Delta = -0.112$  with task-clustered CI  $[-0.129, -0.095]$ ,  $d = -0.91$ . Cells outside the pre-registered family remain descriptive.
- **Multiple-comparison correction: applied to the pre-registered family (2026-07-10).** Holm–Bonferroni correction is applied across the 24 pre-registered Cond-B contrasts (Section 6.1). The wider 9-system by 6-outcome grid still invites 54 pairwise-inference opportunities outside that family; those remain descriptive and uncorrected.
- **Effect-size sensitivity.** Cond-A leads `baseline_3` on `minimum_coverage` by 0.013 (0.981 vs 0.968). This gap is below the pre-registered minimum detectable effect (Cohen’s  $d = 0.25$ ) from Appendix X. Small numerical gaps in the coverage table should be treated as effect-size-sensitive rather than as substantive comparative claims.
- **Independent-lab replication has not been attempted.** All results in this supplement come from the author’s single deposit under the git tags enumerated in Section 8 and Section 10. Independent-lab reruns of the apparatus against a provider-pinned judge stack are not part of the 2026Q2 evaluation. Replication reliability across labs depends on provider version stability (see instrumentation and temporal validity above) and on the executable-package handoff documented in Section 10.

## 11.6 Summary of threats acknowledged versus resolved

Of the nineteen threats enumerated above (three construct, five internal, four external, three instrumentation-and-temporal, four conclusion), three were flagged as structurally resolvable within

v2 additions. Two are now delivered (2026-07-10): bootstrap confidence intervals and Holm–Bonferroni correction for the pre-registered contrast family (Section 6.1). The third, publication of v2 per-judge  $\kappa$  and  $\alpha$  under the shape-neutral rubric, remains open: the v2 20% double-grade sample was not produced before the budget pause, so v2 grading reliability is currently evidenced by the v1 IRR artifact plus the shape-neutral rubric design change, not by a v2-specific reliability measurement. The remaining sixteen threats represent boundary conditions of the evaluation that should be treated as scope statements rather than as flaws: judge-authoring loop, judge-baseline model overlap, field-set anonymization limits, input-condition asymmetry between Cond-A and baselines, cross-vendor LLM fallback disclosure, pretraining-contamination check is one-sided, domain-family scope (all four are governance-family), fine-tune-family scope (Qwen3-only), sample-to-population inference scope, no human-expert coverage baseline, effect-size sensitivity at the Cond-A/baseline\_3 near-tie boundary, rubric-instrument evolution between v1 and v2, baseline authenticity (B4–B6 pattern shells vs literal framework installations), temporal validity (frontier-LLM snapshot drift), construct-scope of subjective coverage outcomes (`mission_intent_match`, `trace_completeness`, `fabrication_count`) as ensemble-mean judge scores, and independent-lab replication uncertainty.

## 12 Reproducibility Checklist

NeurIPS 2025-style reproducibility checklist filled in against this evaluation.

### 1. For all authors.

- **Main claims accurate?** Yes. Five content-tripwire findings across six MANDATE roles, supported by 1480 RunRecords and verbatim quotes.
- **Limitations discussed?** Yes (Section 9).
- **Negative societal impacts?** (Main paper.)
- **Ethics conformance?** (Main paper.) No human-subjects work; all source documents are public-domain or openly-licensed.

### 2. For empirical results.

- **Code and data?** Yes. Zenodo deposit at the DOI listed in `CITATION.cff`.
- **Training and evaluation details?** Yes. Per-role temperatures and `max_tokens` documented in Section 2.3; judge configurations in Section 2.3.
- **Multiple runs?** Yes. 10 runs per (task, system); v1 Phase 8 grading operated on a stratified  $N=700$  sample with 10% IRR under Deviation D-08, which is REVOKED for v2 under Deviation D-10. v2 Stage 4 graded all 12,000 records (Cond-X + Cond-A + Cond-B + B1–B6) under the pre-registered Opus ensemble; the planned 20% double-grade IRR component was *not executed* before the D-13 budget pause and remains an open item (resumable from frozen records). The 9,000 v1 anonymized RunRecords remain on disk as canonical artifacts.
- **Compute and cost?** Phase 6 main matrix:  $\sim 154$  hours Ollama serial for MANDATE-primary +  $\sim 50$ – $80$  hours parallel API for baselines,  $\sim \$447$  API. Phase 8 grading (v1, D-08 sampled):  $\$58.82$  attested +  $\sim \$500$  sunk =  $\sim \$559$  cumulative. v2 projected total under D-10 + D-11 (full pre-registration coverage with O5):  $\sim \$16,181$ , dominated by full-coverage Opus grading ( $\sim \$7,700$ )

and O5 perturbation runs + grading (~\$7,350). The grading components of that projection were priced on a Sonnet-rate per-grade basis; D-10’s restoration of Opus as the Anthropic judge raised per-grade cost  $\sim 5\times$ , and realized multi-provider spend exceeded the projection before Phase B grading completed. Phase B grading was paused at 80.7% coverage under Deviation D-13. Realized spend is tracked in the deposit cost ledger (`engineering_provenance/cost_log/`); per-provider billing dashboards are authoritative for components whose per-record cost fields are not written to disk (see ledger flags).

- **Hardware?** Apple Silicon Mac (M-series), 32 GB+ RAM, Ollama serving six `mandate-*` 32B-parameter fine-tunes.

### 3. For released datasets.

- **Citation and license?** Yes (`CITATION.cff`, `LICENSE`).
- **Responsibly collected?** Yes. NIST SPs, ODNI ATAs, CISA advisories, PCAOB standards, SEC instructions, COSO summary, GAO reports, Twelve-Factor App, Agile Manifesto, GDS Service Standard. All public-domain or openly-licensed.
- **PII/offensive content?** No. Corpus is exclusively organizational and policy documents.

### 4. For models.

- MANDATE-primary’s six role fine-tunes are owned by the upstream MANDATE team and referenced by upstream AEGIS tag `mandate-eval-primary-2026q2-v1`, commit `4f8af83`. Model identifiers recorded in every RunRecord’s `model_versions` field. Full per-role model cards owned by upstream and not redistributed in this artifact.

### 5. For human-subjects work.

- No human subjects. The SME realism audit is scheduled with the v2 full-coverage grading cycle (Section 2.4) and has not yet been executed (open item, Deviation 1).

## 13 Appendix X: Power Analysis for the N=700 Sampled Grading

This appendix documents the statistical power analysis that supports the sample-size choice in Deviation D-08 (Section 9). The choice of N=100 per system across 7 systems was constrained by the Phase 8 budget cap but is independently justified by the analysis below.

### 13.1 Setup

The Phase 8 grading layer produces, per anonymized RunRecord, an ensemble-aggregated score on each of five pre-registered outcomes:

- **O1:** Anchor completeness (fraction of required anchors correctly identified).
- **O2:** Gap detection (true positive rate on planted missing-information probes).
- **O3:** Fabrication rate (proportion of unsupported factual claims; lower is better).
- **O4:** Schema validity (proportion of records that parse against the published JSON schema).

- **O5:** Adversarial resistance (deferred; perturbation suite not run, see Deviation, Section 9).

The cross-system comparison of interest, per outcome, is the mean ensemble score for MANDATE-primary vs. each of the six baselines. With seven systems we have  $\binom{7}{2} = 21$  pair-wise comparisons per outcome; with multiplicity correction the relevant per-test  $\alpha^*$  under Holm-Bonferroni for 4 outcomes (O1-O4) is approximately  $\alpha/k$  for the most-significant test in the family.

### 13.2 Effect Size

The pre-registered minimum detectable effect (MDE) of substantive interest is a standardized mean difference of  $d = 0.25$  (Cohen’s  $d$ , a “small” effect by Cohen’s 1988 convention, but practically meaningful for a binary-grade ensemble score where the typical between-system spread is on the order of  $\sigma \approx 0.20$  in the calibration data).

### 13.3 Power Calculation

For a two-sample  $t$ -test comparing  $n_1 = n_2 = 100$  records per system at  $\alpha = 0.05$  two-tailed, the power to detect  $d = 0.25$  is:

| Quantity                                                  | Value  |
|-----------------------------------------------------------|--------|
| Sample per group ( $n_1 = n_2$ )                          | 100    |
| Standardized effect size ( $d$ )                          | 0.25   |
| Two-tailed $\alpha$ (uncorrected)                         | 0.05   |
| Two-tailed $\alpha^*$ (Holm-Bonferroni, $k = 4$ outcomes) | 0.0125 |
| Power at uncorrected $\alpha = 0.05$                      | 0.42   |
| Power at corrected $\alpha = 0.0125$                      | 0.23   |

The uncorrected power at  $d = 0.25$  is 42%, below the conventional 80% threshold. The corrected family-wise power for the smallest effect we’d care about (across the 4 outcomes simultaneously) is 23%.

However, the comparisons we expect to make at the headline level are not at the  $d = 0.25$  floor; the pilot smoke (Section 6) showed the between-system mean-score spread on schema validity alone is  $\sim 0.60$  (MANDATE-primary  $\sim 0.95$  vs B3 ReAct = 0.000), and on content-tripwire propensities (Section 4) the spread is  $\sim 1.0$  between MANDATE-primary and Sonnet-planner baselines. The same calculation at  $d = 0.5$ :

| Quantity                             | Value |
|--------------------------------------|-------|
| Sample per group                     | 100   |
| Standardized effect size ( $d$ )     | 0.50  |
| Power at uncorrected $\alpha = 0.05$ | 0.94  |
| Power at corrected $\alpha = 0.0125$ | 0.84  |

At  $d = 0.5$  (the headline effect-size range),  $N=100$  per system gives 84–94% power, at or above the conventional 80% threshold.

At  $d = 0.8$  (large effect; the cross-system spread we already see on O4 schema validity from Sections 5.1, 5.4, 10):

| Quantity                             | Value |
|--------------------------------------|-------|
| Sample per group                     | 100   |
| Standardized effect size ( $d$ )     | 0.80  |
| Power at uncorrected $\alpha = 0.05$ | 1.00  |
| Power at corrected $\alpha = 0.0125$ | 0.999 |

(All power values verified via the scipy snippet at the end of this appendix; numbers match to three decimals.)

### 13.4 Confidence Interval Width

A complementary frame is CI width on a single per-system mean score. For  $n = 100$  and a standard deviation of  $\sigma = 0.2$  (typical), the 95% half-width is  $1.96 \cdot \sigma / \sqrt{n} \approx 0.039$ , i.e., per-system mean score reported as  $\hat{\mu} \pm 0.04$ , which is the granularity at which the headline tables would distinguish systems.

### 13.5 Sample-Size Recommendation Summary

- N=100 per system is *under*-powered for the smallest practically-meaningful effect ( $d = 0.25$ , 23% corrected power).
- N=100 per system is *adequately* powered for the headline effect-size range observed in calibration ( $d \geq 0.5$ , 84%+ corrected power).
- N=100 per system produces 95% CI half-widths of  $\pm 0.04$  on per-system mean scores, the granularity at which the cross-system comparison tables would be reported.
- N=100 per system is therefore *sufficient* for the substantive cross-system comparisons reported in the main paper (which are at large effect-size magnitudes), but *insufficient* for hairline small-effect distinctions. The §7 O5 perturbation table reports Wilson 95% CIs on each per-type proportion; the §5.4 headline table reports point estimates; bootstrap CIs and Holm-corrected tests for the pre-registered contrast family are in Section 6.1 (the wider grid remains descriptive). Effect sizes below  $d = 0.25$  are not powered to detect under N=100 per system.

### 13.6 Calculation Provenance

Power values computed under the standard two-sample  $t$ -test non-centrality parameter  $\lambda = d\sqrt{n/2}$ , with degrees of freedom  $\nu = 2n - 2$  and critical value from the inverse- $t$  distribution at  $1 - \alpha^*/2$ . Computation reproducible via:

```
from scipy.stats import nct, t
import math
def power(d, n_per_group, alpha):
    df = 2 * n_per_group - 2
    t_crit = t.ppf(1 - alpha/2, df)
    ncp = d * math.sqrt(n_per_group / 2)
    return 1 - nct.cdf(t_crit, df, ncp) + nct.cdf(-t_crit, df, ncp)

for d in [0.25, 0.50, 0.80]:
    for alpha in [0.05, 0.0125]:
        print(d, alpha, round(power(d, 100, alpha), 3))
```

## 14 References and Cross-Index

The bibliographic references are in the main paper. Cross-index to the `standalone data results/` directory:

- Section 4.1 (Decomposition) → `finding_1_decomposition/` (4 files)
- Section 4.2 (Interpreter) → `finding_2_interpreter/` (4 files)
- Section 4.3 (Validator) → `finding_3_validator/` (3 files)
- Section 4.4 (Binding) → `finding_4_binding/` (3 files)
- Section 4.5 (Intake) → `finding_5_intake/` (3 files)
- Section 5 (Cross-system) → `cross_system/` (2 files)
- Section 3 (Dataset) → `dataset_inventory/` (2 files)

Each finding directory contains:

- `per_record*.jsonl`: searchable per-record entries with `run_id`, `task_id`, the field of interest, and a short context preview. JSONL format: one record per line; queryable with `jq` or any line-oriented tool.
- Markdown files: human-readable verbatim sample sets organized by domain, source document, or other relevant grouping.

The deposit's `DATA_DICTIONARY.md` carries the full schema for every JSON and JSONL field; `CHANGELOG.md` carries the freeze tag chain; `LICENSE` is CC BY 4.0.